

Advancing cancer immunotherapy with protein language models and deep learning IT

Project Thesis

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Abstract

The accurate identification of specific interactions between T-cell receptors (TCRs) and antigenic epitopes is a critical bottleneck in the development of personalized cancer immunotherapies. Current computational methods often struggle to generalize to novel antigens due to the immense theoretical diversity of the immune repertoire and the scarcity of experimentally validated binding data. This thesis addresses these challenges by integrating large-scale pre-trained Protein Language Models (PLMs) with advanced deep learning architectures. Building upon the TEP-NET framework, ProtBERT embeddings are leveraged to capture the evolutionary and physicochemical context of amino acid sequences. A series of architectural enhancements is systematically evaluated, including Symmetric Cross-Attention, Transformer blocks, CNN-based Interaction Maps, and Deep Residual Network (ResNet) classifiers.

The evaluation, conducted under the rigorous TCR-Peptide Pairing (TPP) benchmark, demonstrates that introducing Symmetric Cross-Attention (Model v1) yields the most significant performance gain, achieving a Global ROC AUC of 0.8855 and superior generalization to unseen epitopes (TPP3) compared to the baseline. It is further observed that models trained with Stochastic Gradient Descent (SGD) consistently outperform those trained with ADAM on this biological manifold. Contrasting with trends in other domains, increased architectural complexity via full Transformer blocks resulted in diminishing returns, highlighting the effectiveness of biologically grounded, lightweight bidirectional modeling over heavy over-parameterization. This work confirms that explicitly modeling the mutual inducibility of the TCR-epitope interface is key to advancing predictive immunoinformatics.

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Chapter 1

Introduction

1.1 Initial Situation

Cancer immunotherapy has emerged as a paradigm shift in oncology, leveraging the host’s adaptive immune system to identify and eliminate malignant cells. Central to this process is the recognition of antigenic peptides (epitopes) presented by Major Histocompatibility Complex (MHC) molecules on the surface of tumor cells. This recognition is mediated by T-cell receptors (TCRs), heterodimeric protein complexes capable of distinguishing non-self antigens with high specificity. The success of therapeutic interventions, such as T-cell receptor-engineered T cell (TCR-T) therapies and personalized cancer vaccines, critically depends on the accurate identification of high-affinity TCR-epitope pairs [12]. However, the experimental determination of TCR specificity remains a formidable bottleneck. While high-throughput sequencing has enabled the accumulation of vast repertoires of TCR sequences, the ”ground truth” of binding affinity is typically established through low-throughput, resource-intensive assays such as surface plasmon resonance or X-ray crystallography [3]. Compounding this challenge is the immense theoretical diversity of the TCR repertoire, estimated to exceed 10^{15} possible clonotypes generated via V(D)J recombination. This vast search space creates a severe data scarcity problem: the number of experimentally validated interactions is infinitesimal compared to the universe of possible TCR-pMHC combinations. In recent years, deep learning has been applied to address this gap, formulating binding prediction as a binary classification task. Early architectures, utilizing Convolutional Neural Networks (CNNs) [21] and Recurrent Neural Networks (RNNs) [20], achieved success in classifying interactions involving epitopes seen during training (TPP1 setting). Yet, these models often fail to generalize to novel, unseen epitopes (TPP3/TPP4 settings), which is the primary requirement for clinical applicability in personalized medicine. The reliance on shallow encoding methods (e.g., One-Hot or BLOSUM) limits the ability of these models to capture the complex physicochemical and structural ”grammar” governing molecular docking. Consequently, there is a pressing

need for computational frameworks that can move beyond pattern matching on limited datasets to learning robust, transferable representations of protein interaction.

1.2 Objective

The primary objective of this thesis is to advance the state-of-the-art in TCR-epitope binding affinity prediction by integrating large-scale pre-trained Protein Language Models (PLMs) with advanced deep learning architectures. This thesis builds upon the work of De Gasparo *et al.* [4]. Recognizing that the scarcity of labeled binding data limits the efficacy of training from scratch, this work leverages transfer learning strategies to exploit the structural and evolutionary information implicitly learned by PLMs from massive databases of unlabeled protein sequences. Specifically, this work aims to:

1. **Leverage Contextual Embeddings:** Investigate the utility of dynamic embeddings derived from models such as ProtBERT and ESM-2 to represent TCR and epitope sequences, replacing static physicochemical descriptors.
2. **Enhance Interaction Modeling:** Develop and evaluate a deep learning architecture that utilizes attention mechanisms, specifically Symmetric Cross-Attention and Transformer blocks, to explicitly model the residue-wise interdependencies between the TCR Complementary Determining Region 3 (CDR3) loops and the target epitope.
3. **Improve Generalization:** Address the "unseen epitope" problem by designing a model capable of learning generalized binding rules rather than memorizing specific peptide sequences, thereby improving performance on the rigorous TPP3 and TPP4 evaluation benchmarks.
4. **Integrate Physicochemical Constraints:** Assess whether the fusion of learned PLM representations with explicit physicochemical features (e.g., hydrophobicity, isoelectric point (pI)) yields a synergistic improvement in prediction accuracy.

By combining the representational power of protein language models with structure-aware attention mechanisms, this thesis seeks to contribute to the development of more robust, generalizable tools for computational immunoinformatics.

Chapter 2

Related Work

2.1 CNN Approaches

CNNs, originally designed for computer vision, have been extensively adapted for TCR-epitope binding prediction to capture local dependencies and motif patterns within amino acid sequences. Existing approaches can be broadly categorized by their input representation strategies: those applying 1D convolutions directly to sequence embeddings, those converting interaction pairs into 2D images, and hybrid architectures where CNNs process intermediate feature maps.

Interaction Map and 2D-CNN Approaches

A foundational deviation from sequence-only processing was introduced by ImRex [21], which reframed binding prediction as an image classification problem. Rather than learning separate embeddings for the TCR and epitope, ImRex constructs "Interaction Maps", specifically 2D matrices of size 20×11 representing the pairwise physicochemical differences (hydrophobicity, hydrophilicity, mass, and pI) between amino acids of the CDR3 and the epitope. These maps are processed by a 2D-CNN architecture consisting of filters (3×3) followed by max-pooling and dense layers. This approach posits that spatial patterns in the physicochemical difference matrix correlate with binding affinity, allowing the model to capture interaction interfaces explicitly.

Sequence-Based 1D-CNN Approaches

Conversely, NetTCR-2.0 [20] established that 1D-CNNs applied directly to sequence encodings could achieve competitive performance with computationally lighter architectures. NetTCR-2.0 utilizes a global max-pooling architecture where CDR3 α , CDR3 β , and peptide sequences are encoded via the BLOSUM50 matrix. The model employs 1D convolutional layers with multiple kernel sizes (1, 3, 5, 7, 9) to capture motifs of varying lengths simultaneously. The resulting feature maps are concatenated and passed through dense layers. The authors demonstrated that this "shallow" CNN architecture was sufficient to model the problem complexity given the data constraints at the time, emphasizing the

importance of paired-chain data over architectural depth.

Hybrid and Ensemble Architectures

Recent frameworks have integrated CNNs as components within larger ensemble or hybrid systems to complement attention mechanisms. TSpred [12] employs an ensemble strategy combining a CNN-based module with an attention-based module. The CNN component utilizes 1D convolutions (kernel size 2) on one-hot encoded CDRs and epitope sequences to extract local sequence motifs, which are then combined with the global context captured by reciprocal attention mechanisms. Similarly, TEPCAM [3] integrates CNNs into a Cross-Attention and Multi-channel framework. Unlike ImRex, which builds static physicochemical maps, TEPCAM uses a CNN to process the dynamic attention maps generated by a cross-attention module. These attention maps, representing the learned inter-sequence dependencies between the TCR and epitope, are treated as multi-channel images. A multi-channel CNN is then applied to extract high-level "texture" features from these interaction patterns before the final classification, effectively using convolutions to refine the output of the attention mechanism.

Structural and Generative CNN Applications

CNNs have also been applied to more complex representations involving structural constraints or latent space generation. CATCR [8] introduces a structure-aware framework where CNNs are used to process Residue Contact Matrices predicted by OpenFold. Parallel CNNs extract structural features from these distance matrices, which are then concatenated with sequence features derived from a transformer encoder. This allows the model to leverage spatial proximity information that is lost in pure sequence-based 1D convolutions. Finally, in the domain of unsupervised representation learning, TCR-VALID [16] utilizes CNNs within a Variational Autoencoder framework. The encoder uses lightweight 1D convolutional layers (kernel widths 5, 3, 3) to map physicochemically encoded TCR sequences into a continuous, low-dimensional latent space, while the decoder uses 1D deconvolution layers to reconstruct the sequence. While not a direct binding predictor, this approach demonstrates the utility of CNNs in learning disentangled, continuous representations of the TCR landscape that facilitate clustering and generative tasks.

2.2 Attention Mechanisms and Transformers Approaches

To overcome the limitations of fixed kernel sizes in CNNs, recent architectures have increasingly adopted attention mechanisms and Transformer-based models. These approaches allow for the modeling of long-range dependencies between amino acid residues and the dynamic weighting of interaction sites, regardless of their linear distance in the sequence.

Bimodal and Context-Aware Attention

One of the earlier applications of attention in this domain is TITAN [31], which introduced

a bimodal neural network architecture. Unlike self-attention applied to a single sequence, TITAN employs a "context attention" mechanism. Here, the encoded representation of the epitope acts as a context to compute attention weights for the TCR sequence, and vice versa. This cross-modal attention allows the model to focus on specific regions of the TCR (such as the CDR3 loops) that are most relevant to the physicochemical properties of the specific epitope presented, effectively simulating the induced fit of molecular docking.

Multi-Head Self-Attention Frameworks

Building on the success of the Transformer architecture in natural language processing, ATM-TCR [2] implements a pure multi-head self-attention framework without relying on recurrent or convolutional priors for the initial encoding. The model utilizes separate encoders for the TCR and epitope, where each amino acid's representation is updated based on its interaction with all other residues in the sequence. By employing multiple attention heads, ATM-TCR captures various biological contexts, such as hydrophobicity or electrostatic potential, in parallel. The authors demonstrated that the resulting attention maps are interpretable and correlate with known binding motifs, providing insights into the "binding logic" of the model.

Integration with Pre-trained Language Models

More recent approaches leverage attention not just as an architectural component but as a bridge to integrate high-dimensional embeddings from pre-trained PLMs. EPIC-TRACE [13] combines a convolution-based extraction of local motifs with a multi-head attention mechanism that operates on contextualized embeddings from ProtBERT. This hybrid design allows the model to utilize the rich, structure-aware feature space learned by the PLM while explicitly modeling the pairwise importance of features between the TCR and the epitope. This strategy has proven particularly effective for the challenging task of predicting interactions for unseen epitopes.

Multimodal Attention Encoders

Expanding the scope of input data, MATE-Pred [7] proposes a multimodal attention-based predictor. It treats the TCR and epitope not merely as sequences but as multimodal entities comprising amino acid sequences, physicochemical properties, and structural contact maps (derived from ESM-2). MATE-Pred employs a Transformer-encoder block to fuse these diverse modalities. The self-attention mechanism within the encoder dynamically weighs the contribution of sequence data against structural constraints and physicochemical descriptors, enabling the model to learn a robust, holistic representation of the interaction interface that outperforms unimodal baselines.

2.3 Structure-Aware and Graph-Based Approaches

While sequence-based deep learning models such as NetTCR-2.0 [20] have established strong baselines for TCR-epitope binding prediction, they inherently treat amino acids as

linear strings. This abstraction potentially neglects the crucial three-dimensional spatial constraints and physicochemical interfaces that govern molecular docking. To address this limitation, recent advancements in the field have increasingly incorporated structural priors and graph theory into their architectures.

Graph Neural Networks

GNNs offer a natural representation for protein interactions by modeling amino acids (or atoms) as nodes and their physical or chemical interactions (e.g., covalent bonds, contact distance) as edges. This topology allows models to capture spatial and structural relationships lost in pure sequence-based representations. For instance, the GTE framework [9] constructs a bipartite graph of TCR-epitope interactions, utilizing graph convolution layers to aggregate information from neighboring nodes. This allows the model to learn from the local structural neighborhood of each residue, which is particularly powerful when 3D structural priors are available.

Integration of Structural Priors

Moving beyond graph representations of sequences, some approaches attempt to integrate explicit 3D structural data. TCRcost [17] utilizes 3D-CNNs and Long Short-Term Memory networks (LSTM) to process voxelized representations of the TCR-pMHC complex. Their results demonstrated that correcting predicted structures and utilizing atomic features could improve prediction accuracy compared to pure sequence baselines. Similarly, the GRAPE framework [35] represents a significant leap forward by combining PLMs with graph regularization. By using ESM-2 embeddings to regularize graph attention mechanisms, GRAPE mitigates the "over-smoothing" problem common in GNNs and addresses the severe class imbalance inherent in binding data.

Implicit Structural Learning

However, relying on explicit structure often introduces a computational bottleneck, as generating 3D structures (e.g., via AlphaFold or ESMFold) for millions of sequences is prohibitively expensive. Consequently, recent methods like STAPLER [14] focus on implicit structural learning, demonstrating that transformer-based language models can infer structural motifs through massive pre-training without requiring explicit coordinate files during inference. This thesis builds upon this philosophy, aiming to leverage the structural "grammar" learned by ProtBERT without the computational cost of explicit graph generation.

2.4 Transfer Learning and Pre-training Strategies

The application of deep learning to TCR-epitope binding prediction is often hindered by the scarcity of high-quality, labeled interaction data compared to the vast theoretical diversity of the immune repertoire. To address this, recent approaches have adopted transfer learning strategies, where models are first pre-trained on large-scale unlabeled datasets

to learn the fundamental "grammar" of protein sequences or structural constraints before being fine-tuned on specific binding tasks.

Transformer-based Language Models (BERT adaptations)

Inspired by Natural Language Processing (NLP), several methods utilize the BERT (Bidirectional Encoder Representations from Transformers) architecture to model immune sequences. A key innovation in this space is the adaptation of the Masked Language Modeling (MLM) objective to biological sequences.

TCR-BERT [34] employs a two-stage pre-training strategy to learn a robust representation of T-cell receptors. First, it undergoes a Masked Amino Acid pre-training on approximately 88,000 unlabeled TCR sequences, forcing the model to infer hidden residues based on their context. Unlike generic protein models, TCR-BERT introduces a second, domain-specific pre-training stage focused on antigen classification, using labeled clusters to refine the model's understanding of binding semantics before the final fine-tuning.

Similarly, BERtrand-peptide [22] leverages the BERT architecture but introduces a novel approach to data construction for pre-training. Recognizing the lack of paired data, the authors construct a "hypothetical peptide:TCR repertoire" by randomly pairing experimentally verified MHC-I presented peptides with TCR sequences from healthy donors. This synthetic dataset allows the model to perform MLM on paired inputs, effectively learning cross-sequence dependencies between the TCR and the peptide even before exposure to ground-truth binding labels.

Autoencoder-based Latent Representations

Alternative architectures focus on reconstruction tasks to generate dense numerical embeddings. TEINet [10] utilizes a transfer learning framework based on sequence autoencoders. Instead of a monolithic model, it employs two separately pre-trained encoders (derived from the TCRpeg [11] architecture): one trained on over a million TCR sequences and another on a large library of epitope sequences. These encoders learn to map variable-length sequences into fixed-dimensional latent vectors by minimizing reconstruction error. These pre-trained features are then fed into a feed-forward network, allowing the model to generalize better to unseen epitopes by leveraging the intrinsic features learned during the unsupervised reconstruction phase.

Structure-Aware Protein Language Models

Recent advances have also bridged the gap between sequence modeling and 3D structure prediction. tFold-TCR [33] illustrates how general PLMs can be adapted for immune complex modeling. It builds upon the ESM-2 architecture, extending it to "ESM-PPI-TCR" by further pre-training on a diverse set of protein-protein interactions (PPI), antibodies, and TCR multimers. This specialized pre-training enables the model to extract intra- and inter-chain residue contact information directly from sequence data, bypassing the computationally expensive Multiple Sequence Alignment step typically required for structural modeling. This approach demonstrates that pre-training can capture not just sequence

patterns, but also the evolutionary and structural constraints governing molecular docking.

2.5 Input Representation and Data Handling

The performance of deep learning models in immunoinformatics is heavily contingent upon two design choices: the scope of biological information included in the input representation and the strategy employed to generate artificial negative data (non-binders) for training.

Paired vs. Single Chain Inputs

Early computational approaches predominantly focused on the CDR3 β chain due to the higher availability of beta-chain sequencing data. However, recent benchmarks have demonstrated that this simplification limits prediction accuracy. NetTCR-2.0 [20] provided a quantitative analysis showing that models trained on paired CDR3 α and CDR3 β sequences significantly outperform single-chain baselines, particularly when predicting specificity for pMHC complexes with limited training data. This finding has influenced subsequent architectures like TSpred [12], which mandates paired input to capture the full heterodimeric context of the TCR. Some models, such as EPIC-TRACE [13], extend this scope further by including not just the CDR3 regions but the full variable (V) gene sequences or additional CDR loops (CDR1/2), thereby capturing germline-encoded features that stabilize the pMHC interaction.

Sequential Encodings and RNNs

To handle the variable lengths of amino acid sequences without resorting to manual feature engineering or padding artifacts, RNNs have been widely adopted. ERGO [28] represented a shift from fixed-length physicochemical descriptors to learned representations, utilizing LSTM networks to encode TCR and peptide sequences into dense vectors. By processing sequences step-by-step, ERGO captures sequential dependencies and motifs that static matrices might miss.

Building on this, PiTE [37] employs a two-stage pipeline that integrates pre-training with sequential modeling. Instead of simple one-hot encoding, PiTE utilizes a pre-trained Bidirectional LSTM (BiLSTM), conceptually similar to ELMo [24], to project amino acids into continuous vector spaces based on their context within large unlabeled TCR corpora. While transformer-based methods are gaining traction for global context, PiTE demonstrates that BiLSTMs remain a powerful tool for generating locally contextualized embeddings that outperform static baselines like BLOSUM matrices.

Negative Sampling Strategies

Since experimental assays typically report only positive binding events, the generation of "negative" samples (decoy pairs) is a critical component of the training pipeline. The choice of sampling strategy directly impacts the model's ability to generalize and avoid bias. TEINet [10] provides a comprehensive analysis of these strategies, arguing against

simple random shuffling which can distort the natural frequency distribution of epitopes. Instead, they propose "Unified Epitope Sampling," which preserves the epitope frequency distribution in the negative set, providing a more robust training signal that prevents the model from simply learning to classify based on epitope popularity. Other frameworks adopt biological controls; for instance, MATE-Pred [7] generates negatives by sampling TCR sequences from healthy donor repertoires (presumed non-reactive to the specific target), thereby ensuring that the negative class represents realistic, naturally occurring TCR sequences rather than synthetic permutations.

Chapter 3

Theoretical Foundations

This chapter establishes the fundamental biological mechanisms governing T-cell immunity and provides the theoretical basis for the computational methods employed in this study. It connects the complexity of molecular recognition in immunology with the architectural choices in modern deep learning, specifically focusing on Transformers, Graph Neural Networks (GNNs), and physicochemical feature integration.

3.1 Biological Foundations of TCR-Epitope Recognition

The adaptive immune system relies on the ability of T lymphocytes (T-cells) to distinguish between "self" and "non-self" peptides. This recognition is mediated by the TCR, a heterodimeric protein complex found on the surface of T-cells, which interacts with antigenic peptides (epitopes) presented by MHC molecules [26].

3.1.1 The TCR-pMHC Complex

The interaction involves three key components forming the TCR-pMHC complex:

- **The Epitope:** A short peptide fragment, typically derived from the intracellular degradation of proteins (e.g., viral or tumor antigens). These peptides are displayed on the cell surface by MHC molecules.
- **The MHC Molecule:** A host protein that binds the epitope and presents it to the immune system. The specific variation (allele) of the MHC determines which peptides can be presented, influencing individual susceptibility to diseases [26].
- **The TCR:** The TCR consists mainly of an α - and a β -chain. The binding specificity is primarily determined by the CDRs. Among these, the CDR3 loops (particularly

the CDR3 β) are the most hypervariable and play the most critical role in direct antigen contact [15].

TCR recognition of peptide-MHC complexes relies on structural and physicochemical complementarity combined with dynamic conformational adaptation. TCR-pMHC interactions are governed by induced fit, binding kinetics, and mechanosensitive signaling. When binding persistence and molecular compatibility exceed an activation threshold, intracellular signaling cascades are initiated, ultimately leading to T-cell activation and effector responses such as target cell elimination.

3.1.2 V(D)J Recombination and Repertoire Diversity

A major challenge in modelling this system is the immense diversity of the TCR repertoire. Through a somatic recombination process known as V(D)J recombination, gene segments (Variable, Diversity, and Joining) are randomly rearranged. This stochastic process, combined with random insertions and deletions of nucleotides at the junction sites, results in a theoretical diversity of over 10^{11} possible TCR sequences [32]. This extreme diversity creates a "sparse" data problem for machine learning: the number of experimentally verified interactions is infinitesimal compared to the space of possible combinations, making generalization to unseen TCRs challenging in computational immunology.

3.2 Computational Frameworks for Binding Prediction

Predicting whether a specific TCR binds to a specific epitope is treated computationally as a binary classification problem. Early approaches relied on statistical analysis of sequence similarity, but the field has shifted towards Deep Learning (DL) to capture non-linear dependencies and structural motifs.

3.2.1 Input Representation and Embeddings

To process biological sequences in neural networks, amino acids must be converted into numerical vectors.

- **Physicochemical Properties:** Amino acids can be represented by their intrinsic properties, such as hydrophobicity (Kyte-Doolittle scale), pI, and molecular weight [5]. While useful, relying solely on these features often fails to capture complex sequence context.
- **Learned Embeddings (PLMs):** Modern approaches utilize "embeddings" derived from large language models (e.g., ProtBERT, ESM-2). Similar to how NLP

models learn the context of words in a sentence, these models are pre-trained on millions of protein sequences to learn the "grammar" of amino acids. They project sequences into high-dimensional continuous vector spaces where structurally or functionally similar amino acids are located close to each other [1].

3.2.2 Deep Learning Architectures

CNNs

Originally designed for image processing, CNNs are applied to protein sequences to detect local motifs (e.g., a specific pattern of 3-5 amino acids that facilitates binding). Models like ImRex or the convolutional modules in TEPCAM treat the interaction interface as a 2D map or the sequence as a 1D grid, applying filters to extract local features invariant to their position in the sequence [21].

RNNs and LSTMs

RNNs, and specifically LSTM networks, process sequences step-by-step, maintaining a "memory" of previous residues. This allows them to model sequential dependencies (e.g., the influence of the N-terminus on the C-terminus of a peptide). However, they often struggle with very long-range dependencies and are computationally sequential, limiting parallelization [25].

Transformer Architectures

Transformers have become the state-of-the-art in immunoinformatics (e.g., PiTE, TEP-NET, BERT-based models). Unlike RNNs, Transformers process the entire sequence simultaneously using Self-Attention mechanisms.

- **Self-Attention:** This mechanism allows the model to weigh the importance of every amino acid relative to every other amino acid in the sequence, regardless of their distance. For a TCR, this means the model can learn how a residue at the beginning of the CDR3 loop interacts with one at the end to stabilize the loop structure [15].
- **Cross-Attention:** In the context of binding prediction, cross-attention can explicitly model the interaction between the TCR sequence and the epitope sequence, simulating the "docking" process by attending to relevant parts of the epitope given the TCR context [18].

Graph Neural Networks

GNNs represent the TCR-epitope system as a graph, where amino acids (or atoms) are nodes and their physical or chemical interactions (e.g., covalent bonds, contact distance) are edges. Models like GTE leverage this topology to capture spatial and structural relationships that are lost in pure sequence-based representations. Graph convolution layers (e.g., GraphSAGE) aggregate information from neighboring nodes, allowing the model to learn from the local structural neighborhood of each residue.[6] This is particularly powerful when 3D structural priors (contact maps) are available or can be inferred.

3.2.3 Evaluation Scenarios: The TPP Framework

To rigorously assess a model’s ability to generalize, the TCR-Peptide Pairing framework is often used to categorize test data based on novelty [20]:

- **TPP1 (Seen/Seen):** Both the TCR and the epitope were present in the training set (though not necessarily as a pair). This tests memory/retrieval.
- **TPP2 (Unseen TCR):** The epitope is known, but the TCR is new. This is a common clinical scenario (screening new patients for known viruses).
- **TPP3 (Unseen/Unseen):** Neither the TCR nor the epitope has been seen before. This is the most challenging task, requiring the model to learn universal rules of protein interaction rather than memorizing specific sequences.
- **TPP4 (Unseen Epitope):** The TCR is known, but the epitope is new.

Generalization to TPP3 and TPP4 remains the primary bottleneck for current computational models.

Chapter 4

Methods and Approach

To systematically improve upon the baseline TEP-NET model [4], an experimental framework was adopted that closely mirrors the original data construction while extending the modeling capacity through targeted architectural enhancements. By deliberately preserving the same underlying data sources, class ratios, and train-validation-test partitioning strategy, it was ensured that observed performance differences can be attributed to model design choices rather than dataset discrepancies.

Building on this controlled foundation, a series of neural architectural extensions was designed and evaluated, ranging from single-component augmentations to more complex multi-component configurations. These enhancements include Symmetric Cross-Attention mechanisms, Transformer blocks for contextual sequence modeling, interaction maps implemented via convolutional neural networks, and deep residual classifier heads. This structured progression enables an ablation-style analysis, allowing the contribution of each component to be isolated and assessed under identical data and preprocessing conditions.

The complete methodological pipeline consists of three tightly coupled stages: (i) dataset assembly and stratified splitting following the protocol of De Gasparo *et al.*, [4] (ii) biologically informed preprocessing and feature construction to transform raw amino acid sequences into fixed-dimensional model inputs, and (iii) architectural exploration and model training under controlled class-balancing constraints. The following sections describe these stages in detail.

4.1 Dataset

For comparability with prior work, particularly the dataset construction in De Gasparo *et al.*, [4] the training, validation, and test datasets were assembled from the same publicly available TCR-epitope sequence resources. Positive TCR-epitope pairs were aggregated from VDJdb [27], ImmuneCODE [23], and pMTnet [19], which together provide experimentally validated TCR specificity information across diverse antigenic con-

texts. VDJdb is a curated repository of antigen-specific TCR sequences that has been actively maintained since its initial releases and continues to incorporate new specificity records from the literature and community submissions through at least mid-2025. The ImmuneCODE dataset, originally released in 2020 and updated through Release 002.x by 2025, contains hundreds of millions of TCR sequences and a large subset of high-confidence SARS-CoV-2-associated TCR–epitope bindings. The pMTnet dataset provides additional high-confidence TCR–epitope pairings and has been widely used as a benchmark in TCR–peptide binding research.

After downloading and merging positive interaction data from these sources, sequence cleansing was performed to ensure consistency and duplicates were removed such that each unique TCR–epitope pair appeared only once. For the training and validation sets, only pairs meeting quality thresholds comparable to those described in De Gasparo *et al.* [4] were retained, resulting in a training set of 61 638 total positive examples composed of 58 655 unique TCRs and 847 unique epitopes, and a validation set of 12 328 positive examples. The test set was drawn from held-out sources including IEDB [30], McPAS-TCR [29], PDB [36] complex annotations, and TBAdB [38] entries not present in the training/validation splits, yielding 21 282 positive examples with 20 037 unique TCRs and 630 unique epitopes.

Negative samples for all subsets were derived from the 10xGenomics experimental dataset, which provides a large repertoire of paired TCR sequences without known epitopes. From this source, 6 453 181 negative TCR–epitope pairs comprising 129 591 unique TCR sequences and 50 unique epitopes were extracted. Negative sampling was performed in a manner analogous to De Gasparo *et al.* [4] in order to balance representation across epitope space.

To ensure consistent class balance with prior works, the validation and test sets were constructed with a positive:negative ratio of 1:5, matching the ratios used in De Gasparo *et al.* [4] for these subsets. Splits were further stratified across the four TPP categories (TPP1–TPP4) to assess generalization under increasingly challenging conditions, i.e., from seen TCRs and epitopes in new combinations (TPP1) to both unseen TCRs and unseen epitopes (TPP3).

In summary, the dataset integrates the latest versions of these community standards as available in June of 2025 and, by adopting the same positive–negative ratios and partitioning strategies, facilitates direct comparison with existing predictive models evaluated on analogous benchmarks.

4.2 Preprocessing

The raw biological data required extensive preprocessing to transform variable-length amino acid sequences into fixed-dimensional numerical tensors suitable for the hybrid

deep learning architecture. This pipeline was divided into offline feature generation and online data handling during training.

4.2.1 Sequence Encoding and Dimensionality Reduction

To capture the complex biochemical context of the immune receptors, the ProtBERT protein language model was utilized. Unlike static encoding schemes (e.g., One-Hot), ProtBERT generates dynamic embeddings where the representation of each amino acid is influenced by its surrounding sequence context.

- **Embedding Generation:** Full TCR CDR3 β sequences and epitope sequences were passed through the pre-trained ProtBERT model, yielding a 1024-dimensional vector for each amino acid residue.
- **Dimensionality Reduction (PCA):** To reduce the memory footprint and computational load while preserving the most salient variance in the data, PCA was applied to the raw embeddings. The 1024-dimensional vectors were projected down to 64 principal components. This creates a dense, information-rich representation for each residue.

4.2.2 Sequence Standardization (Padding and Truncation)

As shown in the model architectures, the neural network requires fixed input shapes. The following sequence length constraints were established, based on the distribution of lengths in the dataset:

- **TCR CDR3:** Fixed to a length of **26 residues**.
- **Epitope:** Fixed to a length of **24 residues**.

Sequences exceeding these limits were truncated, and sequences shorter than these thresholds were zero-padded to match the tensor shapes (26, 64) and (24, 64), respectively.

4.2.3 Physicochemical Feature Engineering

To explicitly model the biochemical compatibility between the TCR and the epitope, a set of **12 explicit physicochemical descriptors** was calculated. Unlike the latent features learned by ProtBERT, these features provide the model with direct information regarding molecular stability and reactivity. The selected features are:

- **Kidera Factors (KF1, KF7):** Descriptors correlating with helix/bend preference (KF1) and flat/extended preference (KF7), capturing structural propensities.

- **Hydrophobicity and Aromaticity:** Crucial for analyzing the hydrophobic core interactions common in protein binding interfaces.
- **Isoelectric Point:** Determining the charge state of the peptide at physiological pH.
- **Instability Index:** Estimating the stability of the peptide in vitro.

These values were computed for both the TCR and the epitope, resulting in a global feature vector of dimension 12 (6 features $\times$ 2 sequences).

4.2.4 Dynamic Data Balancing

Given the extreme class imbalance in immunological data (where non-binders vastly outnumber binders), training on the raw distribution would lead to model bias. A dynamic, batch-level balancing strategy within the data loader was implemented. Instead of down-sampling the negative class permanently, a custom generator that constructs training batches with a strictly enforced 50/50 ratio of positive to negative samples was employed. For every batch of size N , the generator randomly samples $N/2$ verified binders and $N/2$ non-binders (sampled from the larger pool of real negatives), ensuring the model receives a balanced gradient signal throughout the training process.

4.3 Baseline Architecture (v0)

The baseline model (v0) establishes the fundamental framework for this study (see figure 4.1), replicating and adapting the architecture proposed in the original TEP-NET thesis. It is designed as a multi-modal neural network that processes three distinct inputs: the TCR sequence embedding, the Epitope sequence embedding, and the vector of physicochemical features.

Input Processing: The model accepts pre-computed, PCA-reduced ProtBERT embeddings. The TCR input is a tensor of shape (26, 64), and the Epitope input is a tensor of shape (24, 64). Additionally, a vector of 12 physicochemical features is passed through a custom embedding layer (either Piecewise Linear Encoding or Periodic Embeddings) to map continuous variables into a learnable high-dimensional space.

Asymmetric Attention Mechanism: A defining characteristic of the baseline is its asymmetrical processing of the interaction. The core interaction module is a Multi-Head Attention layer where the TCR sequence serves as the *Query*, while the Epitope sequence serves as both the *Key* and *Value*. This design implies a directional dependency, modeling how the TCR "attends" to specific regions of the epitope.

Classification Head: The output of the attention mechanism is subjected to Global Average Pooling (1D) to derive a fixed-length vector. This vector is concatenated with the

flattened physicochemical embeddings. The combined representation is passed through a dense classification block consisting of a fully connected layer with N units (hyperparameter `ff_dim`), followed by a Dropout layer for regularization. The final output is generated by a single neuron with a Sigmoid activation function, interpreting the result as a binding probability $P(y = 1|x)$.

4.4 Architectural Improvements

To systematically address potential bottlenecks in the baseline, specifically its directional bias and shallow depth, four distinct architectural enhancements were developed (see figure 4.2, which presents the architecture of model version 8 with all enhancements combined).

4.4.1 Symmetric Cross-Attention (v1)

Biological binding is a mutual physicochemical event; the epitope’s conformation influences the TCR just as the TCR recognizes the epitope. The baseline’s unidirectional attention (TCR querying Epitope) potentially discards information regarding how the epitope “perceives” the TCR.

Model *v1* introduces **Symmetric Cross-Attention** (SCA). In addition to the original attention branch, a parallel Multi-Head Attention layer is introduced where the Epitope embedding acts as the *Query* and the TCR embedding acts as the *Key* and *Value*. The pooled outputs of both branches ($Attention_{T \rightarrow E}$ and $Attention_{E \rightarrow T}$) are concatenated before the classification stage. This ensures the model captures a holistic view of the binding interface, effectively doubling the interaction bandwidth.

4.4.2 Transformer Blocks (v2)

The baseline utilizes raw Multi-Head Attention layers, which can suffer from training instability in deeper networks. Modern Sequence-to-Sequence models typically encapsulate attention within robust **Transformer Blocks** (TB).

In Model *v2* (and variants *v6*, *v7*, *v8*), the raw attention layers are replaced by custom Transformer Blocks. Each block consists of two sub-layers:

1. A Multi-Head Attention mechanism followed by Dropout, a Residual Connection (Add), and Layer Normalization.
2. A position-wise Feed-Forward Network (FFN) consisting of two dense layers (expansion and projection) with ReLU activation, followed by Dropout, a Residual Connection, and Layer Normalization.

This structure allows the model to not only attend to interaction sites but also to process and refine the extracted features non-linearly within the interaction module itself.

4.4.3 Interaction Map with 2D CNN (v3)

Sequence-based attention models implicitly learn relationships, but they do not explicitly represent the pairwise compatibility matrix often used in structural biology (e.g., contact maps). Model *v3* introduces an explicit **Interaction Map** (IM) module.

The outer product (dot product) of the TCR embedding $T \in \mathbb{R}^{26 \times d}$ and Epitope embedding $E \in \mathbb{R}^{24 \times d}$ was computed along the feature dimension. This results in a 2D spatial grid $M \in \mathbb{R}^{26 \times 24 \times 1}$, where each cell $M_{i,j}$ represents the cosine similarity between the i -th residue of the TCR and the j -th residue of the epitope. This "image" of the interaction is processed by a 2D CNN comprising two convolutional layers (32 and 64 filters, 3×3 kernels) followed by Global Max Pooling. This branch is designed to detect local binding motifs, such as diagonal patterns indicative of aligned hydrophobic patches.

4.4.4 Deep ResNet Classifier Head (v4)

The baseline employs a shallow classification head (single dense layer), which limits its capacity to model complex, non-linear relationships between the extracted interaction features and the physicochemical properties.

Model *v4* replaces the simple dense layers with a **Deep ResNet Classifier** (RNH). This module consists of a stack of L blocks, where each block contains:

- A Dense layer ($dim = ff_dim$) with non-linear activation.
- A Dropout layer.
- A second Dense layer ($dim = ff_dim$).
- A second Dropout layer.
- A Residual Connection adding the input of the block to the output.
- Layer Normalization.

This ResNet structure allows the network to act as a deeper function approximator without suffering from the vanishing gradient problem, facilitating the learning of higher-order abstractions from the input features.

4.5 Combinatorial Architectures

To investigate whether these improvements are synergistic or redundant, models representing the power set of these features were implemented:

- **v5 (SCA + IM + RNH):** Combines Symmetric Attention, the CNN-based Interaction Map, and the Deep ResNet Head.
- **v6 (SCA + TB + IM):** Combines Symmetric Attention, Transformer Blocks (replacing raw attention), and the Interaction Map.
- **v7 (SCA + TB + RNH):** Combines Symmetric Attention, Transformer Blocks, and the Deep ResNet Head.
- **v8 (Full Architecture):** The most complex variant, integrating all components: Symmetric Attention utilizing Transformer Blocks, parallel Interaction Map processing, and the Deep ResNet Classifier.

4.6 Hyperparameter Tuning and Training Protocol

To ensure a fair comparison, each model version ($v1 - v8$) underwent a rigorous hyperparameter optimization process. The **Optuna** framework was utilized to conduct a Bayesian search over the hyperparameter space. The base model uses the optimal hyperparameters proposed by the TEP-NET thesis.

Tuning Strategy: Two distinct tuning runs were performed for each model, one utilizing the **Stochastic Gradient Descent** optimizer and one utilizing the **ADAM** optimizer. Each tuning session consisted of 35 trials. The objective function maximized the Area Under the Receiver Operating Characteristic Curve (ROC AUC) on the validation set.

Optimized Parameters: The search space included:

- Batch Size (16, 32, 64, 128)
- Learning Rate ($1e^{-4}$ to $1e^{-2}$, log scale)
- Dropout Rate (0.0 to 0.5)
- L2 Regularization ($1e^{-5}$ to $1e^{-2}$)
- Feed-Forward Dimension (**ff_dim**): Integer range [16, 200]
- Number of Layers (1 to 5)

- Number of Attention Heads (2 to 50)
- Activation Function (ReLU, Tanh, Leaky ReLU)
- Physicochemical Embedding Type (PLE vs. Periodic)

Following the identification of the optimal hyperparameters, the models were fully trained on the combined training set. Training utilized class-balanced batches (generated dynamically) to mitigate the 1:5 imbalance in the raw data, ensuring the model saw an equal number of binders and non-binders during gradient updates.

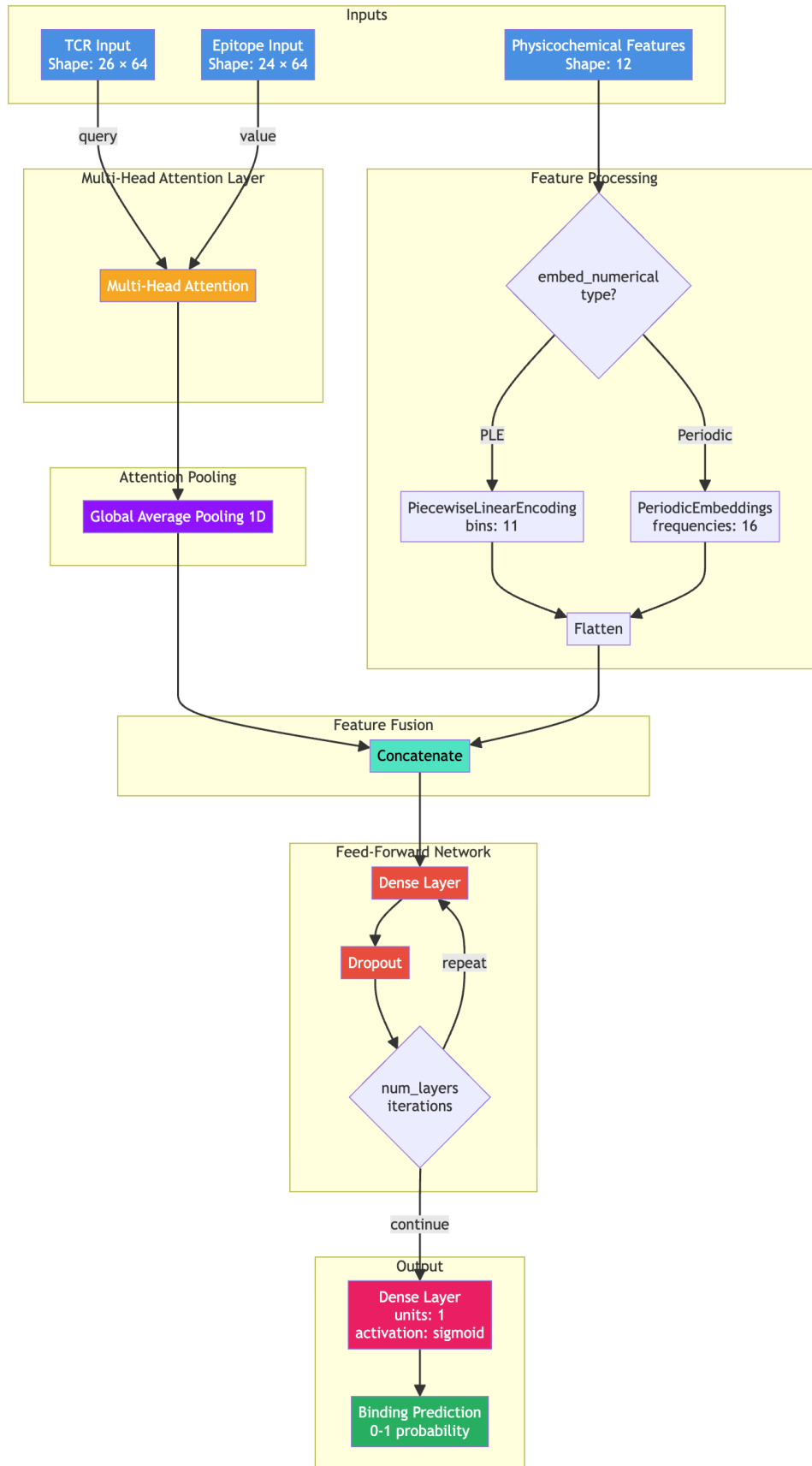


Figure 4.1: Architecture of Baseline Model (v0).

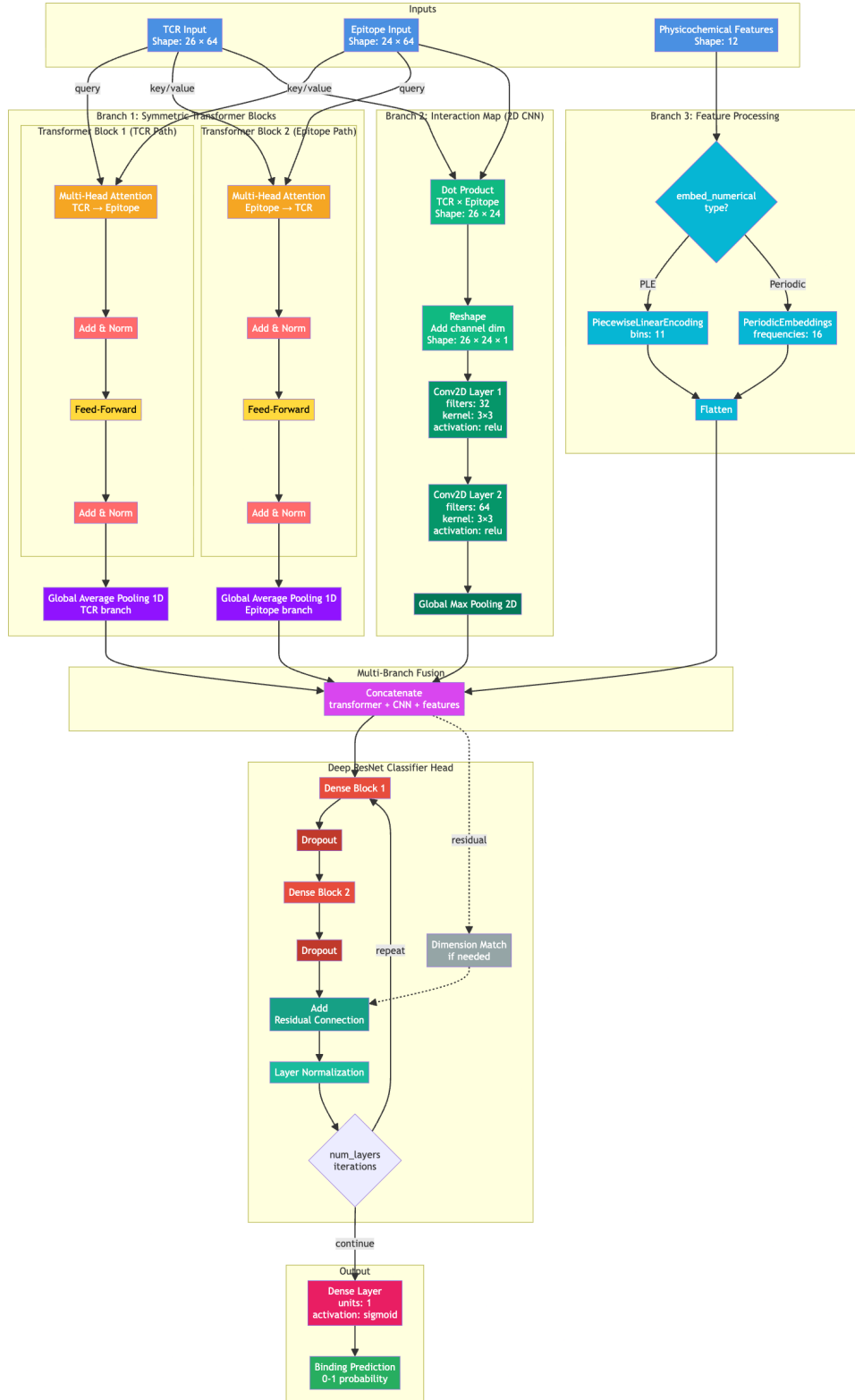


Figure 4.2: Architecture of Model Version 8 using Symmetric Cross-Attention, Transformer Blocks, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head.

Chapter 5

Results

The models were evaluated using the TPP framework, which assesses generalization across four distinct categories of increasing difficulty: TPP1 (Seen pairs), TPP2 (Unseen TCR), TPP3 (Unseen TCR and Epitope), and TPP4 (Unseen Epitope).

5.1 Optimizer Performance: SGD vs. ADAM

A decisive trend was observed regarding the optimization algorithm. Contrary to standard deep learning practices where ADAM often yields faster convergence, the experiments conducted in this work demonstrated that models trained with **SGD** consistently outperformed those trained with ADAM across almost all relevant metrics (see figure 5.1).

For the best-performing architecture (*v1*), the use of SGD resulted in a Global ROC of **0.8855**, compared to just **0.7379** with ADAM. Similarly, the Global F1-score dropped from 0.6521 (SGD) to 0.3756 (ADAM). This trend suggests that for this specific manifold of biological data, ADAM may converge to sharp local minima that fail to generalize, whereas SGD finds flatter, more robust minima. Consequently, the subsequent detailed analysis focuses exclusively on the results obtained using the SGD optimizer.

5.2 Model Performance Evaluation

5.2.1 The Superiority of Symmetric Attention (*v1*)

The introduction of Symmetric Cross-Attention (*v1*) yielded the most significant performance gains. Model *v1* achieved the highest scores across the majority of metrics, validating the hypothesis that bidirectional information flow is critical for binding prediction.

- **Global Performance:** *v1* achieved a Global ROC of **0.8855** and an F1-Score of **0.6521**, significantly outperforming the Baseline (*v0*) ROC of 0.8494 and F1 of

0.5998.

- **Generalization:** Crucially, *v1* demonstrated robust generalization in the hardest category, TPP3 (Unseen TCR/Unseen Epitope), achieving an F1-score of **0.7513** compared to the baseline’s 0.7288.

5.2.2 Deep ResNet Heads and Generalization (v4)

Model *v4* (Symmetric Attention + Deep ResNet Head) presented a compelling trade-off. While its Global F1-score (0.4800) was lower than the baseline, it exhibited exceptional performance in specific generalization tasks.

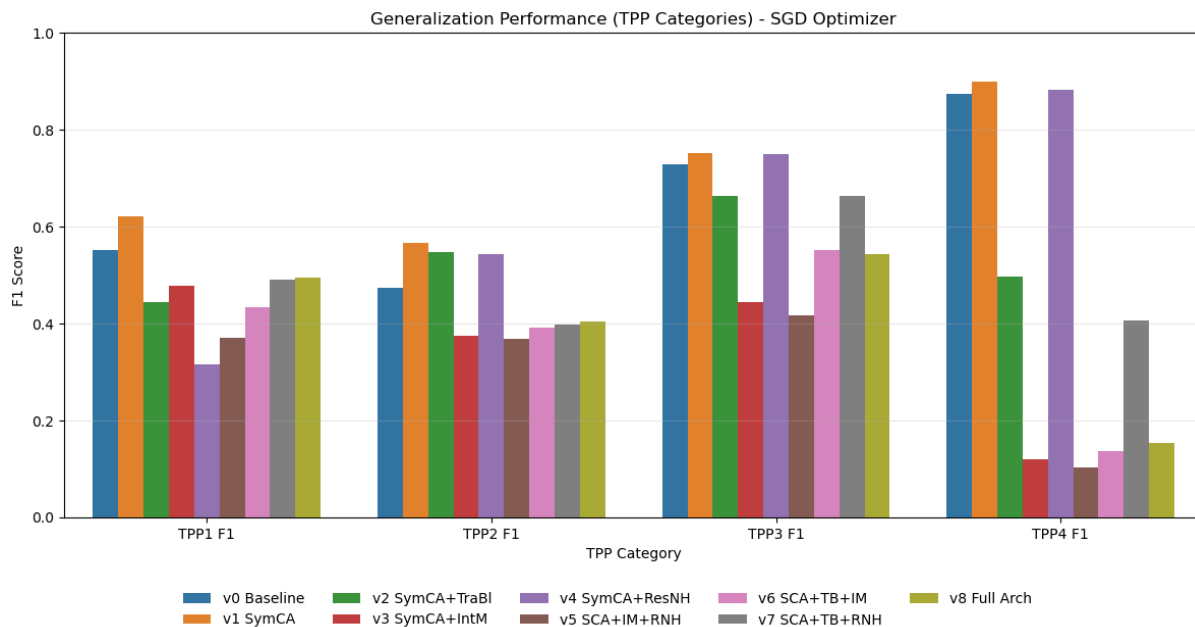
- *v4* achieved the highest recorded score for **TPP3 (0.7494)** and **TPP4 (0.8818)**, surpassing the baseline’s TPP4 score of 0.8736.
- However, it suffered in TPP1 (0.3153 vs Baseline 0.5513), suggesting that the deep residual layers force the model to learn abstract rules rather than memorizing training examples, which is a desirable trait for discovering novel binders.

5.2.3 Degradation in Complex Architectures

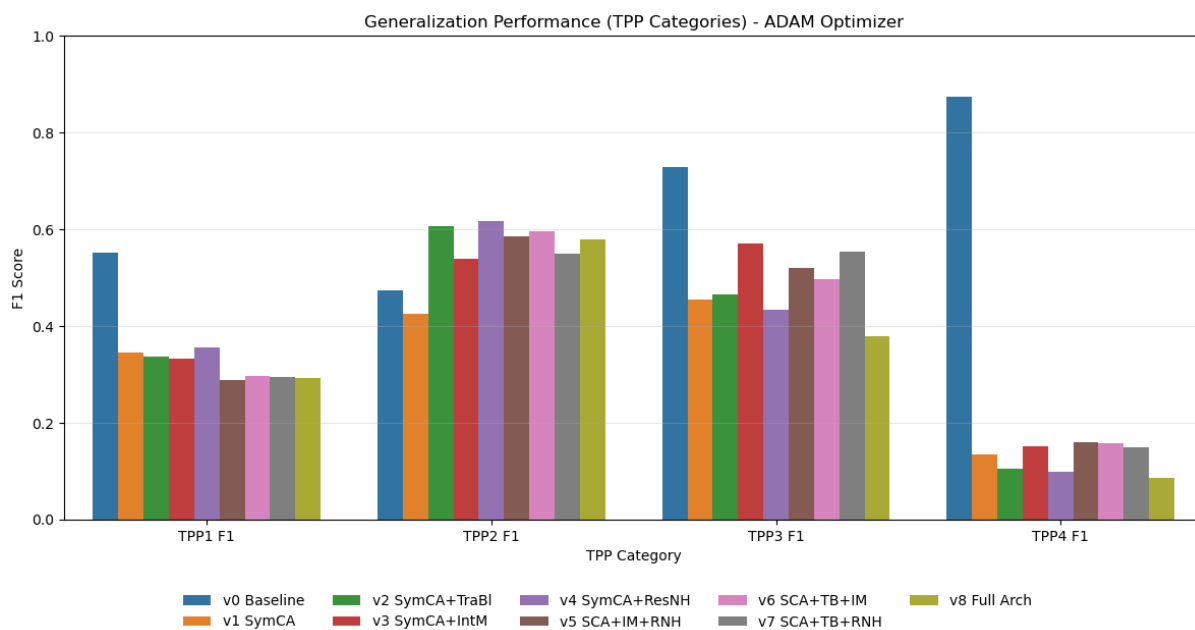
Unexpectedly, increasing architectural complexity lead to diminishing returns and performance degradation.

- **Transformer Blocks (v2, v6-v8):** Models incorporating full Transformer blocks generally underperformed. For example, the Full Architecture (*v8*) achieved a Global ROC of only 0.7988, falling well behind the simpler *v1*.
- **Interaction Maps (v3, v5):** The addition of the CNN-based Interaction Map (*v3*) resulted in a sharp drop in Global ROC to 0.7461. This suggests that the pairwise dot-product features might be redundant to the attention mechanism or require significantly more data to train effectively.

In summary, the simplest structural enhancement (symmetry) provided the most robust improvement, while deeper classifier heads offered a pathway to better generalization at the cost of precision (see figures 5.2, 5.3).



(a) using the SGD optimizer



(b) using the ADAM optimizer

Figure 5.1: TPP performance optimizer comparison.

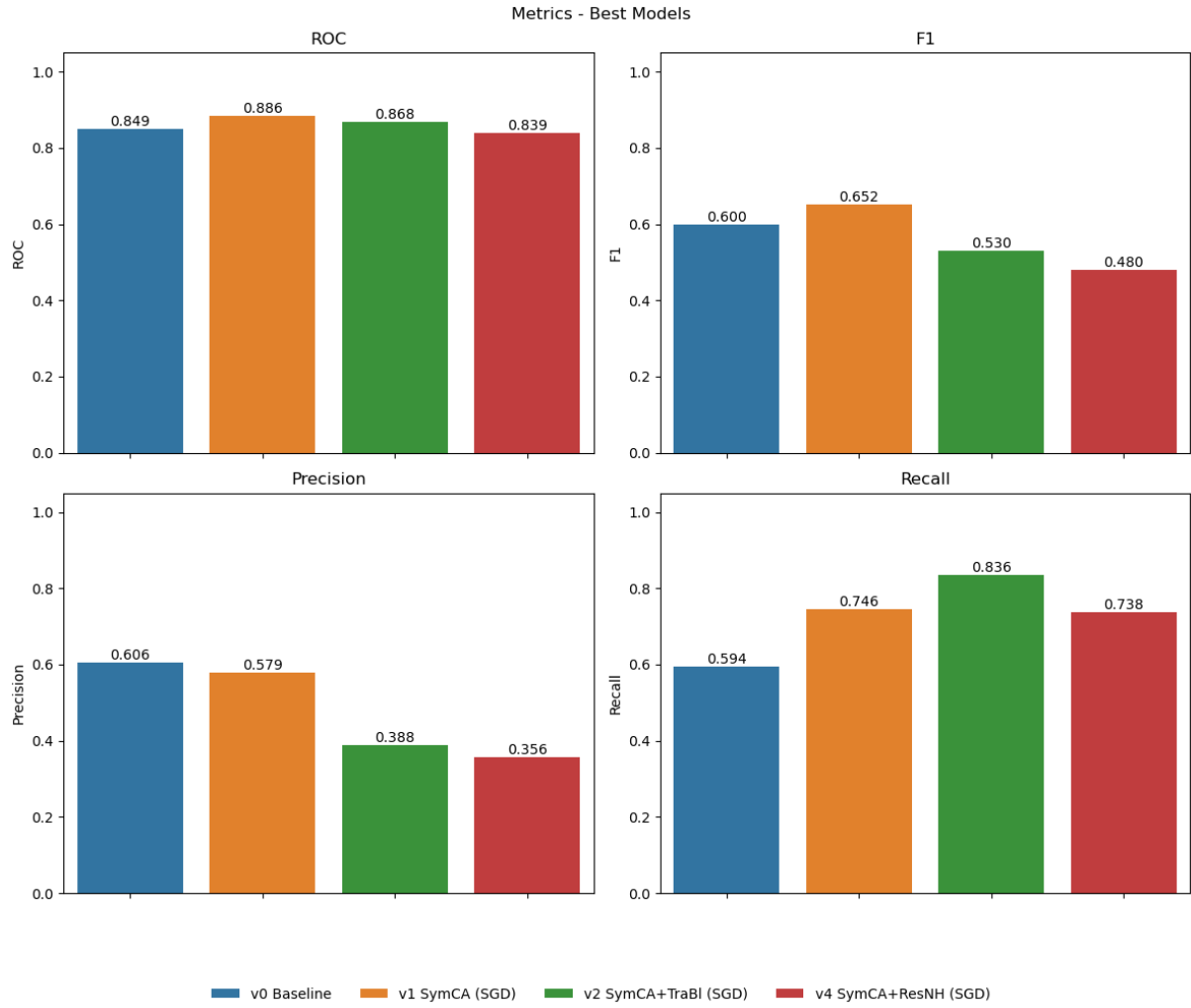


Figure 5.2: Metrics Plot of best Models.

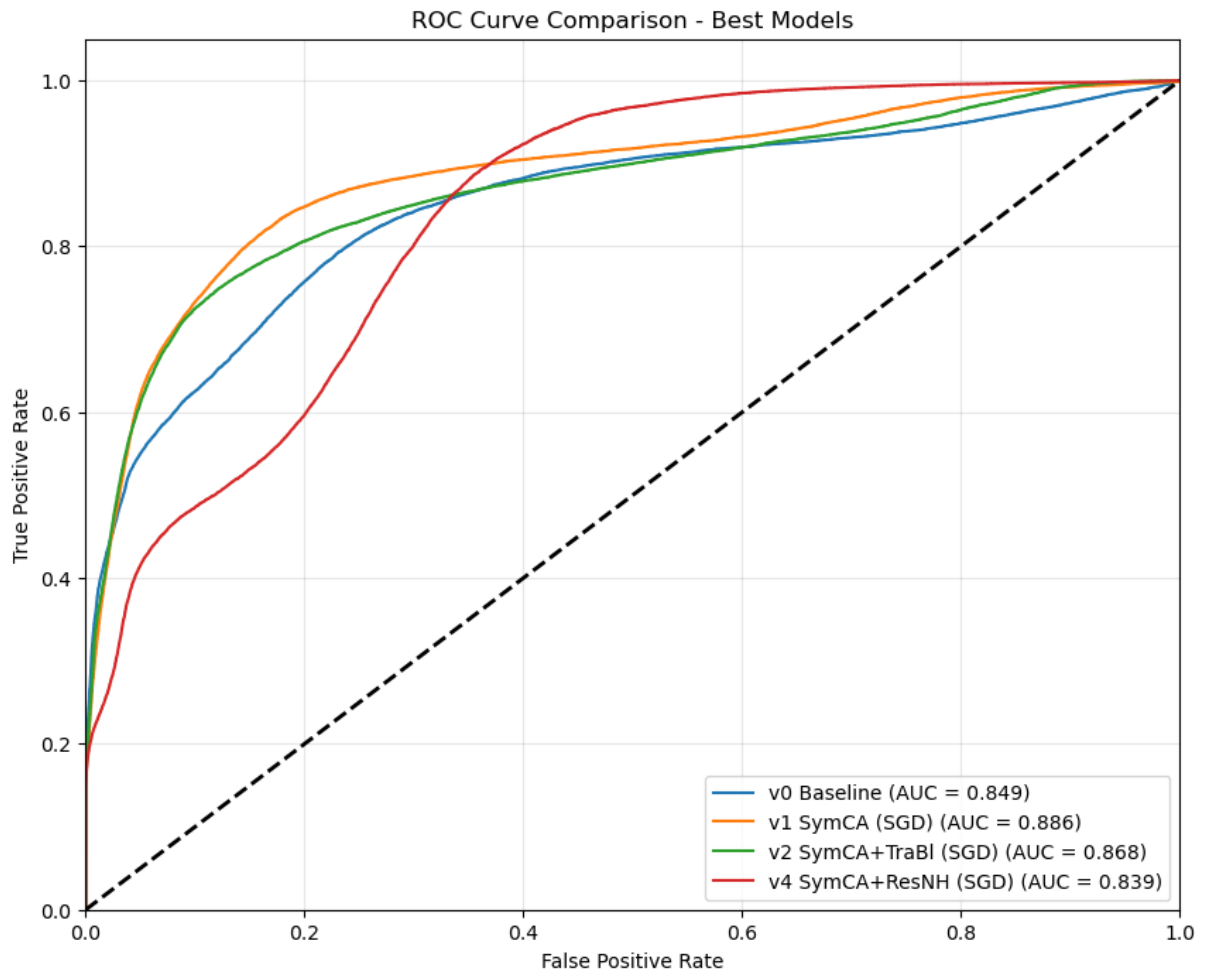


Figure 5.3: ROC Curves of best Models.

Chapter 6

Discussion and Outlook

6.1 The Effectiveness of Symmetric Architectures

The standout performance of Model *v1* validates the biological intuition that TCR-epitope binding is a bidirectional dependency. In the baseline TEP-NET (and many standard transformer models), the attention mechanism is unidirectional: the TCR queries the epitope. This treats the epitope as a static context. By introducing Symmetric Cross-Attention, the model is allowed to explicitly calculate how the epitope "queries" the TCR. This architectural symmetry doubles the informational bandwidth of the interaction module without significantly increasing the parameter count, leading to the universal improvements observed in ROC AUC and TPP3 scores. It implies that for biological pair-classification tasks, mutual inducibility is a key feature to model.

6.2 Generalization vs. Memorization (*v4*)

The results from Model *v4* (Deep ResNet Head) highlight a classic machine learning dichotomy: memorization versus generalization. The baseline model (*v0*) performed well on TPP1 (Seen pairs), likely relying on memorizing specific sequence motifs. Model *v4*, equipped with a deep residual classifier, performed poorly on TPP1 but excelled in TPP4 (Unseen Epitope). We hypothesize that the depth of the ResNet head acts as an information bottleneck that filters out superficial sequence memorization, forcing the model to rely on higher-order physicochemical abstractions. This makes *v4* a potentially superior candidate for clinical applications where the goal is to screen patients for entirely novel viral epitopes, even if it appears less accurate on known benchmarks.

6.3 Failure of Complex Models: The Shared Dimension Bottleneck

A critical finding of this thesis is the failure of the "Full Architecture" (*v8*) and Transformer-based variants to outperform the simpler models. While over-parameterization and data scarcity likely played a role, a retrospective analysis of the codebase reveals a structural limitation in the implementation of the Transformer Blocks.

In standard Transformer architectures (e.g., BERT), the internal FFN is expanded to a dimension $d_{ff} = 4 \times d_{model}$ to allow for feature disentanglement. However, in this implementation, the hyperparameter `ff_dim` was shared between the Transformer's internal FFN and the final classifier's dense layers. During hyperparameter tuning, the optimizer frequently selected lower values for `ff_dim` (e.g., 47 or 64) to regularize the classifier head. This optimization inadvertently "choked" the Transformer blocks, forcing their internal projection layers to be extremely narrow. Consequently, the Transformer blocks in models *v2*, *v6*, *v7*, and *v8* were likely operating with insufficient capacity to learn complex representations, rendering them less effective than the raw, unconstrained attention mechanism used in *v1*.

6.4 Outlook and Future Work

Based on these findings, future research directions should focus on:

1. **Decoupling Hyperparameters:** The tuning process must be refined to separate the `ff_dim` of the Transformer blocks from the `ff_dim` of the classifier head. This would allow the Transformer to expand features as intended while keeping the classifier compact to prevent overfitting.
2. **Unsupervised Pre-training:** To make the complex Interaction Map (CNN) and Transformer architectures viable, future work should explore unsupervised pre-training on unpaired TCR and Epitope databases. Initializing the weights via Masked Language Modeling on domain-specific data could provide the necessary priors to train these larger models effectively on the limited binding dataset.

In conclusion, this study demonstrates that in the domain of TCR-epitope prediction, biologically grounded architectural simplicity, specifically symmetric attention, currently yields the highest returns, though deep residual networks offer a promising avenue for improving generalization to novel antigens.

6.5 Future Outlook: Geometric Deep Learning for Interaction Prediction

The results presented in this thesis confirm that while sequence-based embeddings from large PLMs capture significant evolutionary semantics, they may lack the explicit geometric context required to distinguish between subtle binding variants. Biological binding is fundamentally a function of shape complementarity and electrostatics in 3D space, features that are often compressed or lost when flattening a protein into a 1D sequence representation.

6.5.1 The Geometric Graph Neural Network Approach

To bridge this representational gap, future work should prioritize the development of Geometric Graph Neural Networks. In this proposed architecture, the TCR and epitope are represented not as linear sequences, but as spatial graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. In this topology, nodes $\mathcal{V}$ represent amino acids (centered on C_α atoms) and edges $\mathcal{E}$ represent physical proximity (e.g., Euclidean distance $< 8\text{\AA}$).

By utilizing architectures such as Graph Attention Networks v2 (GATv2), the model can dynamically learn to "attend" to structurally critical residues, specifically those in the CDR3 loop that physically contact the epitope, while effectively down-weighting non-interacting framework regions. This creates a "structure-aware" attention map that inherently provides explainability, allowing researchers to visualize binding hotspots directly on the molecular surface rather than inferring them from sequence attention weights.

6.5.2 Feasibility via Hybrid Structural Acquisition

A major historical barrier to this approach has been the scarcity of experimental crystal structures for TCR-pMHC complexes. However, the advent of ultra-fast folding models like ESMFold now makes it feasible to generate ("hallucinate") structures for millions of sequences with high confidence.

A proposed pipeline for future research involves a **Hybrid Structural Acquisition Strategy**:

1. **AlphaFold Database (AFDB) Querying:** Instantly retrieving high-quality, pre-computed structures for known viral epitopes using their UniProt IDs.
2. **On-the-fly Folding:** Utilizing ESMFold explicitly for the hypervariable, unseen TCR sequences where database matches are unavailable.

By combining these hallucinated structures with explicit physicochemical node features (e.g., Kidera factors, hydrophobicity), future models could move beyond pattern

matching amino acid strings to learning the underlying "physics of binding." This geometric inductive bias holds the potential to solve the generalizability issues currently plaguing the field, particularly in the challenging TPP3 and TPP4 settings.

Bibliography

- [1] Nadav Brandes, Dan Ofer, Yam Peleg, Nadav Rappoport, and Michal Linial. Proteinbert: a universal deep-learning model of protein sequence and function. *Bioinformatics*, 2022.
- [2] Michael Cai, Seo-Jin Bang, Pengfei Zhang, and Heewook Lee. Atm-tcr: Tcr-epitope binding affinity prediction using a multi-head self-attention model. *Frontiers in Immunology*, 2022.
- [3] Junwei Chen, Bowen Zhao, Shenggeng Lin, Heqi Sun, Xueying Mao, Meng Wang, Yanyi Chu, Liang Hong, Dong-Qing Wei, Min Li, and Yi Xiong. Tepcam: Prediction of t-cell receptor–epitope binding specificity via interpretable deep learning. *Protein Science*, 2024.
- [4] Sven De Gasparo and Jasmina Bogojeska. Tep-net: A deep neural network for tcr-epitope binding prediction using physicochemical and embedding features. *CAI ZHAW*, 2024.
- [5] Donard Dwyer. Amino acids: Chemical properties. *Wiley Encyclopedia of Chemical Biology*, 2008.
- [6] Alex Fout, Jonathon Byrd, Basir Shariat, and Asa Ben-Hur. Protein interface prediction using graph convolutional networks. 2017.
- [7] Etienne Goffinet, Raghvendra Mall, Ankita Singh, Rahul Kaushik, and Filippo Castiglione. Mate-pred: Multimodal attention-based tcr-epitope interaction predictor. *bioRxiv*, 2024.
- [8] Hongchen Ji, Xiang-Xu Wang, Qiong Zhang, Chengkai Zhang, and Hong-Mei Zhang. Predicting tcr sequences for unseen antigen epitopes using structural and sequence features. *Briefings in Bioinformatics*, 2024.
- [9] Feng Jiang, Yuzhi Guo, Hehuan Ma, Saiyang Na, Wenliang Zhong, Yi Han, Tao Wang, and Junzhou Huang. Gte: a graph learning framework for prediction of t-cell receptors and epitopes binding specificity. *Briefings in Bioinformatics*, 2024.

-
- [10] Yuepeng Jiang, Miaoze Huo, and Shuai Cheng Li. Teinet: a deep learning framework for prediction of tcr-epitope binding specificity. *bioRxiv*, 2022.
 - [11] Yuepeng Jiang and Shuai Cheng Li. Deep autoregressive generative models capture the intrinsics embedded in t-cell receptor repertoires. *Briefings in Bioinformatics*, 2022.
 - [12] Ha Young Kim, Sungsik Kim, Woong-Yang Park, and Dongsup Kim. Tspred: a robust prediction framework for tcr-epitope interactions using paired chain tcr sequence data. *Bioinformatics*, 2024.
 - [13] Dani Korpela, Emmi Jokinen, Alexandru Dumitrescu, Jani Huuhtanen, Satu Mustjoki, and Harri Lähdesmäki. Epic-trace: predicting tcr binding to unseen epitopes using attention and contextualized embeddings. *Bioinformatics*, 2023.
 - [14] Bjørn P. Y. Kwee, Marius Messemaker, Eric Marcus, Giacomo Oliveira, Wouter Scheper, et al. Stapler: Efficient learning of tcr-peptide specificity prediction from full-length tcr-peptide data. *bioRxiv*.
 - [15] Esteban Lanzarotti, Paolo Marcatili, and Morten Nielsen. T-cell receptor cognate target prediction based on paired α and β chain sequence and structural cdr loop similarities. *Frontiers in Immunology*, 2019.
 - [16] Allen Y. Leary, Darius Scott, Namita T. Gupta, Janelle C. Waite, Dimitris Skokos, Gurinder S. Atwal, and Peter G. Hawkins. Designing meaningful continuous representations of t cell receptor sequences with deep generative models. *bioRxiv*, 2023.
 - [17] Fan Li, Xinyang Qian, Xiaoyan Zhu, Xin Lai, Xuanping Zhang, and Jiayin Wang. Tercost: a deep learning model utilizing tcr 3d structure for enhanced of tcr-peptide binding. *Frontiers in Genetics*, 2024.
 - [18] Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. 2019.
 - [19] Tianshi Lu, Ze Zhang, James Zhu, Yunguan Wang, Peixin Jiang, Xue Xiao, Chantale Bernatchez, John Heymach, Don Gibbons, Jun Wang, Lin Xu, Alexandre Reuben, and Tao Wang. Deep learning-based prediction of the t cell receptor-antigen binding specificity. *Nature Machine Intelligence*, 2021.
 - [20] Alessandro Montemurro, Viktoria Schuster, Helle Rus Povlsen, Amalie Kai Bentzen, Vanessa Jurtz, et al. Nettcr-2.0 enables accurate prediction of tcr-peptide binding by using paired tcr α and β sequence data. *Communications Biology*, 2021.

-
- [21] Pieter Moris, Joey De Pauw, Anna Postovskaya, Sofie Gielis, Nicolas De Neuter, Wout Bittremieux, Benson Ogunjimi, Kris Laukens, and Pieter Meysman. Current challenges for unseen-epitope tcr interaction prediction and a new perspective derived from image classification. *Briefings in Bioinformatics*, 2020.
- [22] Alexander Myronov, Giovanni Mazzocco, Paulina Król, and Dariusz Plewczynski. Bertrand—peptide:tcr binding prediction using bidirectional encoder representations from transformers augmented with random tcr pairing. *Bioinformatics*, 2023.
- [23] Sean Nolan, Marissa Vignali, Mark Klinger, Jennifer Dines, Ian Kaplan, Emily Svejnoha, Tracy Craft, Katie Boland, Mitchell Pesesky, Rachel Gittelman, Thomas Snyder, Christopher Gooley, Simona Semprini, Claudio Cerchione, Fabio Nicolini, Massimiliano Mazza, Ottavia Delmonte, Kerry Dobbs, Gonzalo Carreño, and Harlan Robins. A large-scale database of t-cell receptor beta sequences and binding associations from natural and synthetic exposure to sars-cov-2. *Frontiers in Immunology*, 2025.
- [24] Matthew Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. Deep contextualized word representations. 2018.
- [25] Thirupathi Reddy. Drawbacks of lstm algorithm: A case study. 2025.
- [26] Anna-Lisa Schaap-Johansen, Milena Vujović, Annie Borch, Sine Reker Hadrup, and Paolo Marcatili. T cell epitope prediction and its application to immunotherapy. *Frontiers in Immunology*, 2021.
- [27] Mikhail Shugay, Dmitriy V Bagaev, Ivan V Zvyagin, Renske M Vroomans, Jeremy Chase Crawford, Garry Dolton, Ekaterina A Komech, Anastasiya L Sycheva, Anna E Koneva, Evgeniy S Egorov, Alexey V Eliseev, Ewald Van Dyk, Pradyot Dash, Meriem Attaf, Cristina Rius, Kristin Ladell, James E McLaren, Katherine K Matthews, E Bridie Clemens, Daniel C Douek, Fabio Luciani, Debbie van Baarle, Katherine Kedzierska, Can Kesmir, Paul G Thomas, David A Price, Andrew K Sewell, and Dmitriy M Chudakov. Vdjdb: a curated database of t-cell receptor sequences with known antigen specificity. *Nucleic Acids Research*, 2017.
- [28] Ido Springer, Hanan Besser, Nili Tickotsky-Moskovitz, Shirit Dvorkin, and Yoram Louzoun. Prediction of specific tcr-peptide binding from large dictionaries of tcr-peptide pairs. *Frontiers in Immunology*, 2020.
- [29] Nili Tickotsky, Tal Sagiv, Jaime Prilusky, Eric Shifrut, and Nir Friedman. Mcpas-tcr: a manually curated catalogue of pathology-associated t cell receptor sequences. *Bioinformatics*, 2017.

- [30] Randi Vita, Nina Blazeska, Daniel Marrama, IEDB Curation Team Members, Sebastian Duesing, Jason Bennett, Jason Greenbaum, Marcus De Almeida Mendes, Jarjapu Mahita, Daniel K Wheeler, Jason R Cantrell, James A Overton, Darren A Natale, Alessandro Sette, and Bjoern Peters. The immune epitope database (iedb): 2024 update. *Nucleic Acids Research*, 2024.
- [31] Anna Weber, Jannis Born, and María Rodríguez Martínez. Titan: T-cell receptor specificity prediction with bimodal attention networks. *Bioinformatics*, 2021.
- [32] Nan-Ping Weng. Numbers and odds: Tcr repertoire size and its age changes impacting on t cell functions. *Seminars in immunology*, 2023.
- [33] Fandi Wu, Yu Zhao, Yang Xiao, Chenchen Qin, Fang Wang, Zihan Wu, Long-Kai Huang, Xiao Liu, Jiangning Song, Bing He, Jamie Rossjohn, and Jianhua Yao. Fast and accurate modeling of tcr-peptide-mhc complexes using tfold-tcr. *bioRxiv*, 2025.
- [34] Kevin Wu, Kathryn E. Yost, Bence Daniel, Julia A. Belk, Yu Xia, Takeshi Egawa, Ansuman Satpathy, Howard Y. Chang, and James Zou. Tcr-bert: learning the grammar of t-cell receptors for flexible antigen-xbinding analyses. *bioRxiv*, 2021.
- [35] Haowen Chen Xiangzheng Fu, Li Peng et al. Grape: graph-regularized protein language modeling unlocks tcr-epitope binding specificity. *Briefings in Bioinformatics*, 2025.
- [36] Christine Zardecki, Shuchismita Dutta, David S. Goodsell, Robert Lowe, Maria Voigt, and Stephen K. Burley. Pdb-101: Educational resources supporting molecular explorations through biology and medicine. *Protein Science*, 2021.
- [37] Pengfei Zhang, Seo-Jin Bang, and Heewook Lee. Pite: Tcr-epitope binding affinity prediction pipeline using transformer-based sequence encoder. *Pacific Symposium on Biocomputing*, 2022.
- [38] Wei Zhang, Longlong Wang, Ke Liu, Xiaofeng Wei, Kai Yang, Wensi Du, Shiyu Wang, Nannan Guo, Chuanchuan Ma, Lihua Luo, Jinghua Wu, Liya Lin, Fan Yang, Fei Gao, Xie Wang, Tao Li, Ruifang Zhang, Nitin K Saksena, Huanming Yang, Jian Wang, Lin Fang, Yong Hou, Xun Xu, and Xiao Liu. Pird: Pan immune repertoire database. *Bioinformatics*, 2019.

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List of Abbreviations

Abbreviation	Full Word
TCR	T-Cell Receptor
TPP	TCR-Peptide Pairing
MHC	Major Histocompatibility Complex
CDR	Complementary Determining Region
pI	Isoelectric Point
PCA	Principal Component Analysis
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
GNN	Graph Neural Network
FFN	Feed-forward Network
DL	Deep Learning
PLM	Protein Language Model
MLM	Masked Language Modeling
SGD	Stochastic Gradient Descent
KF	Kidera Factor
LSTM	Long Short-Term Memory
SymCA, SCA	Symmetric Cross-Attention
TraBl, TB	Transformer Block
IntM, IM	Interaction Map
ResNH, RNH	Deep ResNet Classifier Head

Appendix A

Additional Materials

A.1 Model Architecture Diagrams

A.2 Model Performance Metrics Tables

A.3 Model Performance Metrics Plots

Model	Global ROC	Global F1	Global Prec	Global Rec	TPP1 F1	TPP2 F1	TPP3 F1	TPP4 F1
v0 Baseline	0.849442	0.599796	0.605585	0.594117	0.551284	0.474802	0.728782	0.873610
v1 SymCA	0.885537	0.652064	0.579405	0.745560	0.621036	0.566739	0.751264	0.899209
v2 SymCA+TraBl	0.868464	0.530030	0.388026	0.835965	0.443478	0.547787	0.663242	0.496241
v3 SymCA+IntM	0.746111	0.405274	0.288098	0.683112	0.478737	0.375314	0.445182	0.120029
v4 SymCA+ResNH	0.838869	0.480042	0.355775	0.737713	0.315264	0.544283	0.749377	0.881764
v5 SCA+IM+RNH	0.700460	0.378775	0.326248	0.451461	0.372014	0.387111	0.441254	0.166295
v6 SCA+TB+IM	0.788930	0.428347	0.295113	0.780895	0.434674	0.392630	0.551839	0.136543
v7 SCA+TB+RNH	0.823094	0.511833	0.456884	0.581806	0.490422	0.398964	0.662628	0.407273
v8 Full Arch	0.798806	0.456220	0.339660	0.694578	0.495614	0.404874	0.543180	0.154213

Table A.1: Performance Metrics utilizing SGD Optimizer

Model	Global ROC	Global F1	Global Prec	Global Rec	TPP1 F1	TPP2 F1	TPP3 F1	TPP4 F1
v0 Baseline	0.849442	0.599796	0.605585	0.594117	0.551284	0.474802	0.728782	0.873610
v1 SymCA	0.737940	0.375606	0.267535	0.630157	0.344581	0.425141	0.454009	0.135395
v2 SymCA+TraBl	0.750493	0.405574	0.261486	0.903346	0.333485	0.609121	0.465830	0.105815
v3 SymCA+IntM	0.820380	0.421753	0.270994	0.950569	0.329385	0.522147	0.568518	0.152481
v4 SymCA+ResNH	0.773676	0.404610	0.258523	0.930317	0.355437	0.616166	0.433512	0.099790
v5 SCA+IM+RNH	0.739964	0.400672	0.255791	0.924067	0.287951	0.585203	0.519822	0.160931
v6 SCA+TB+IM	0.796937	0.404187	0.262337	0.880039	0.297615	0.595565	0.496405	0.157410
v7 SCA+TB+RNH	0.763346	0.406543	0.274045	0.787097	0.280483	0.576458	0.565054	0.156635
v8 Full Arch	0.674219	0.349803	0.217185	0.898365	0.295662	0.578750	0.378903	0.085596

Table A.2: Performance Metrics utilizing Adam Optimizer

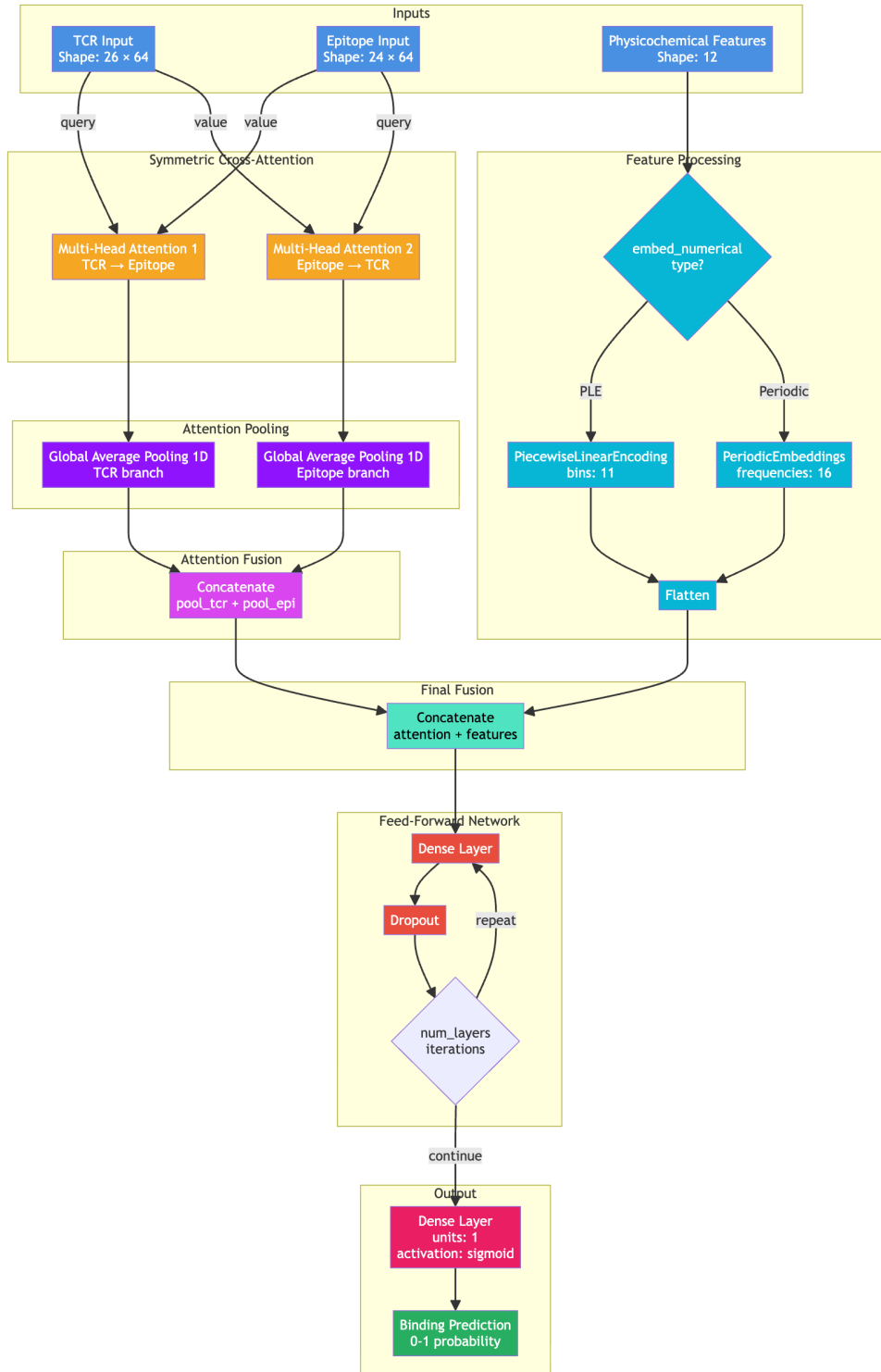


Figure A.1: Architecture of Model Version 1 using Symmetric Cross-Attention.

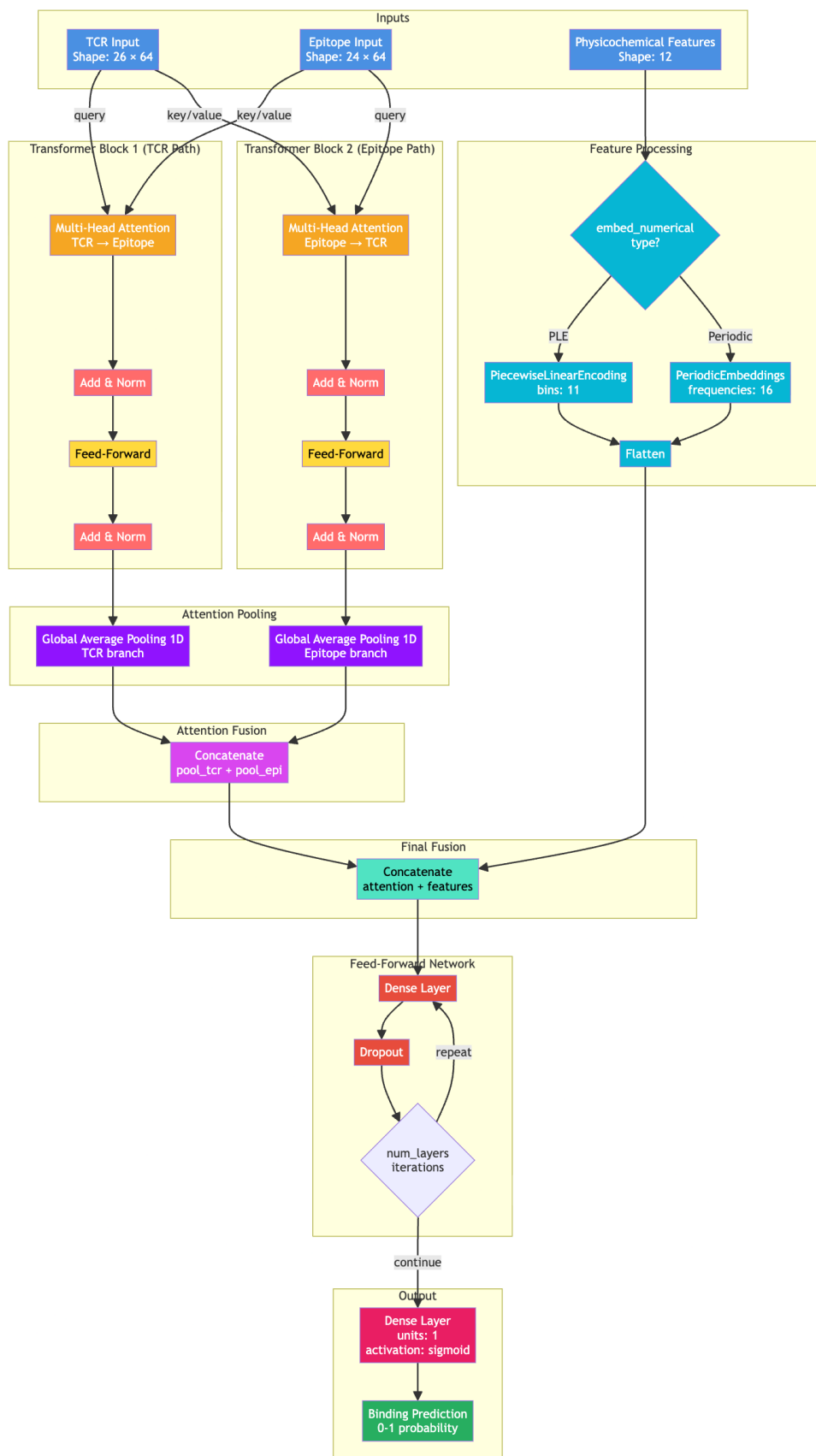


Figure A.2: Architecture of Model Version 2 using Symmetric Cross-Attention and Transformer Blocks.

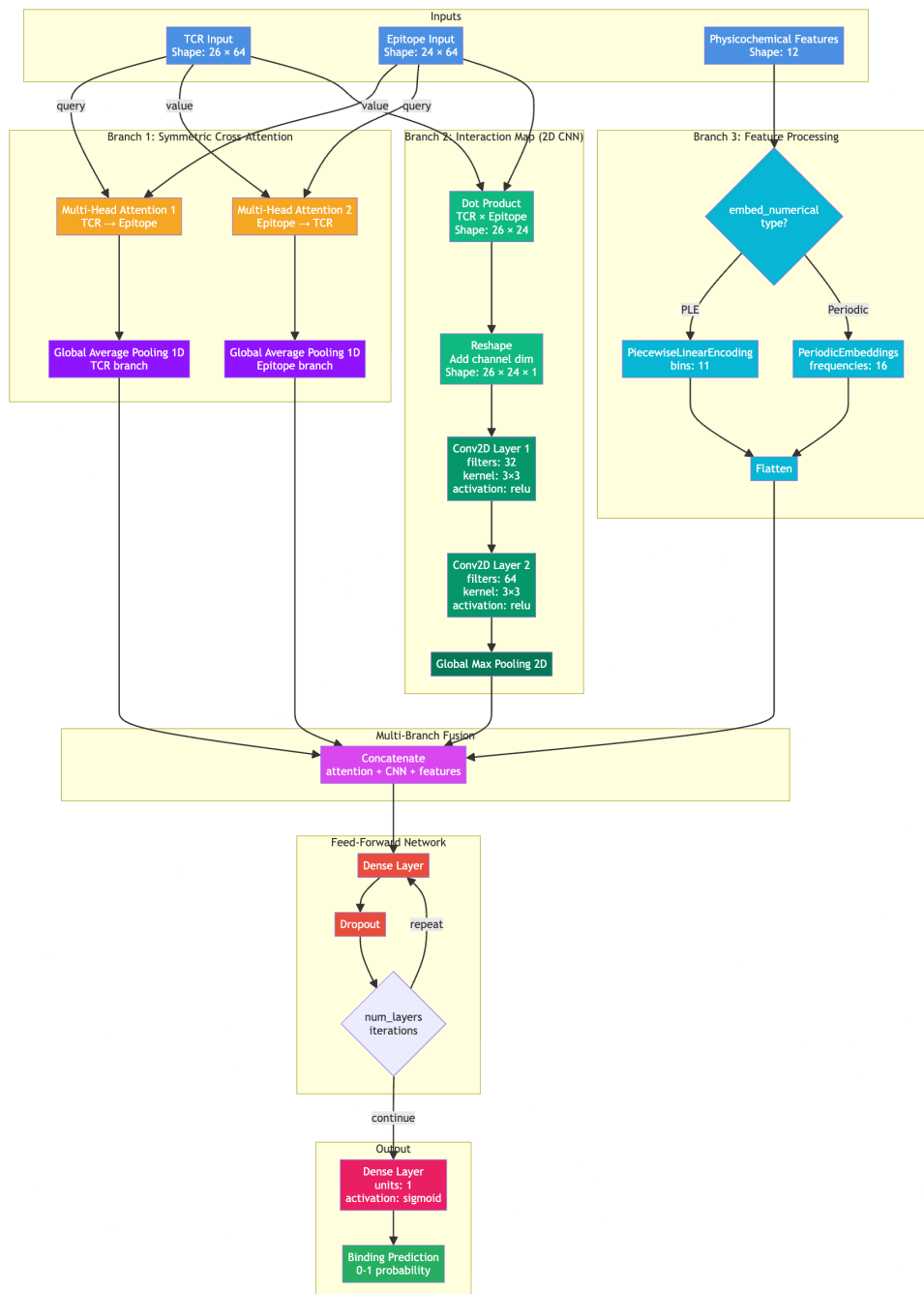


Figure A.3: Architecture of Model Version 3 using Symmetric Cross-Attention and an Interaction Map (2D CNN).

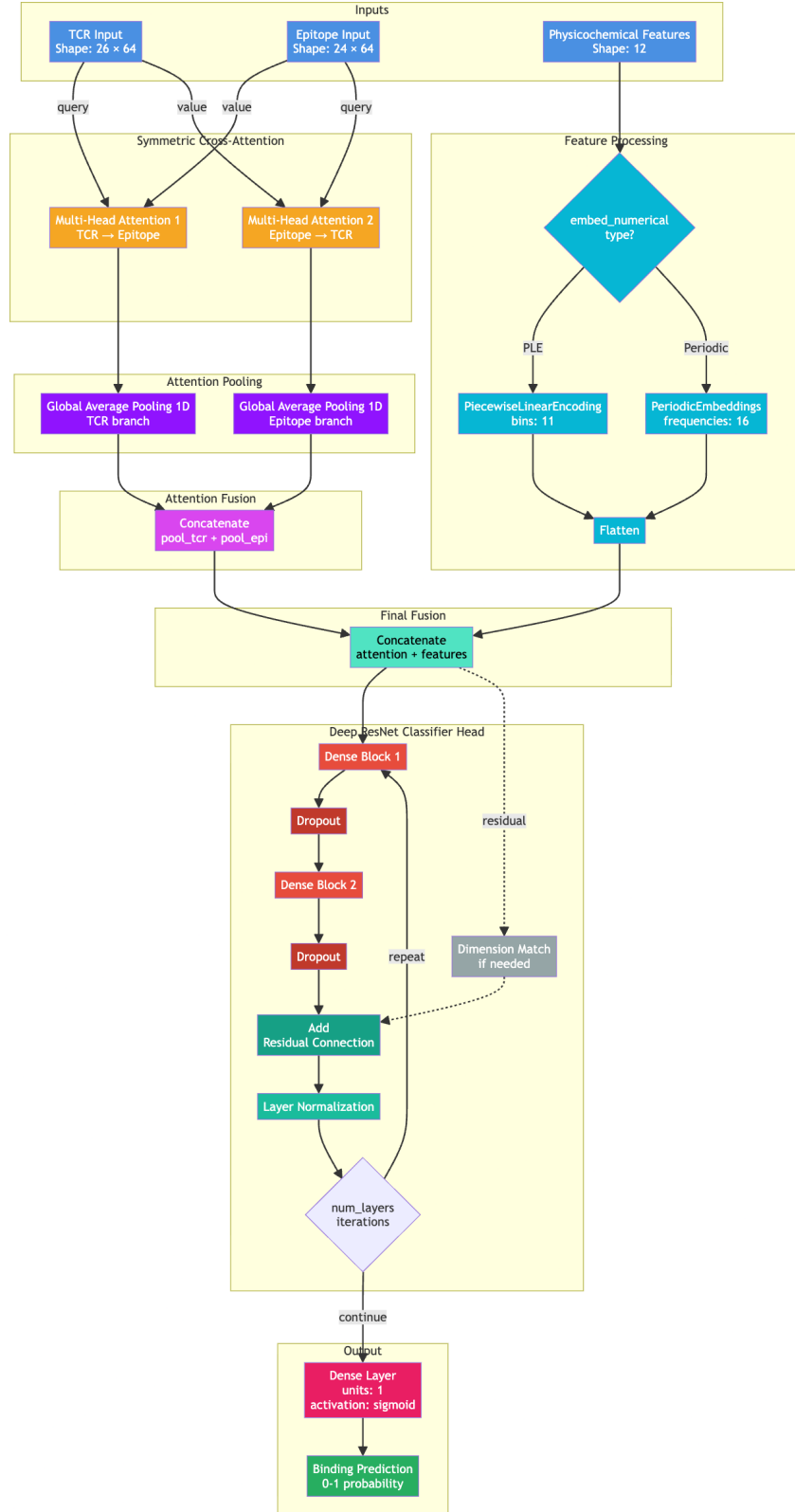


Figure A.4: Architecture of Model Version 4 using Symmetric Cross-Attention and a Deep ResNet Classifier Head.

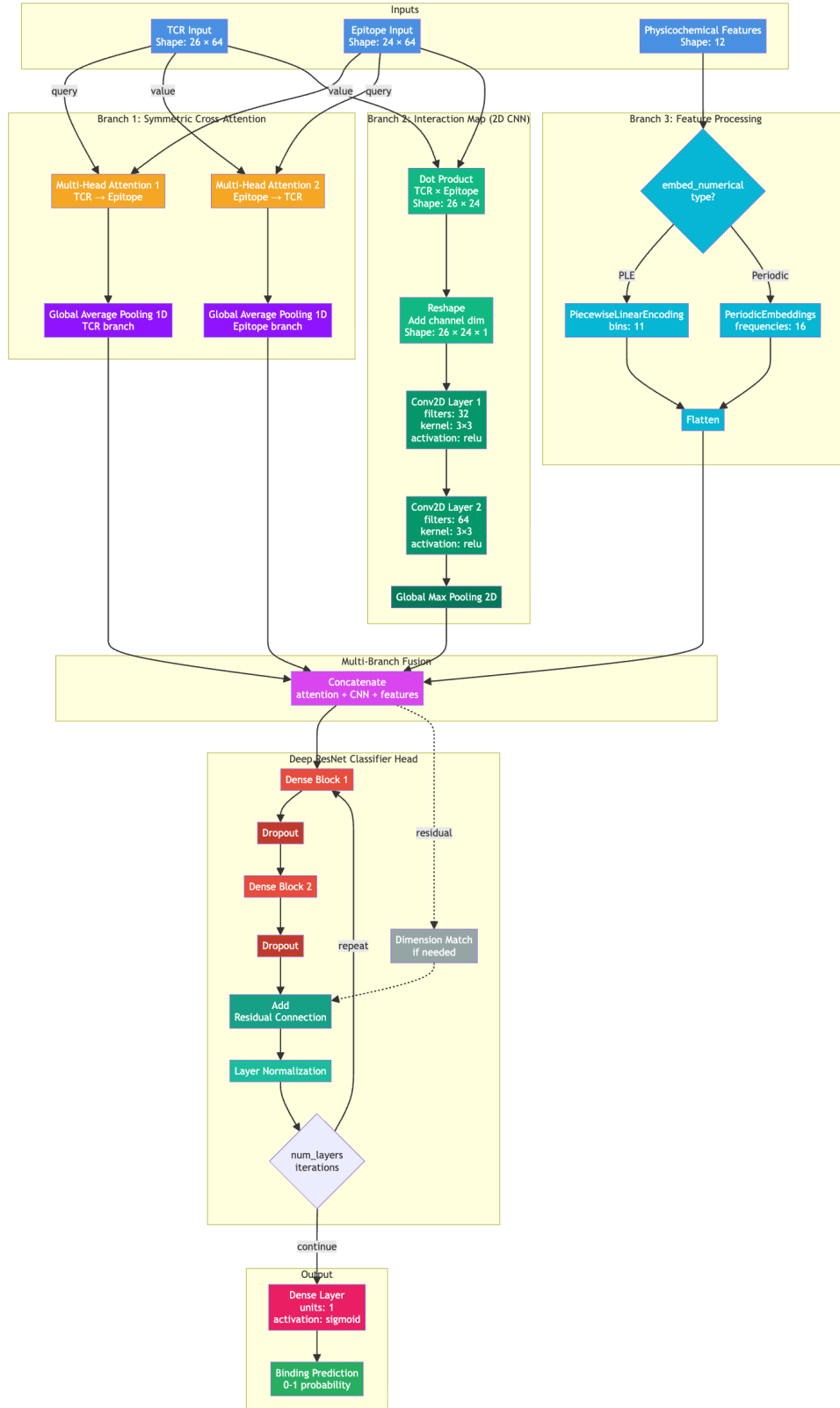


Figure A.5: Architecture of Model Version 5 using Symmetric Cross-Attention, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head.

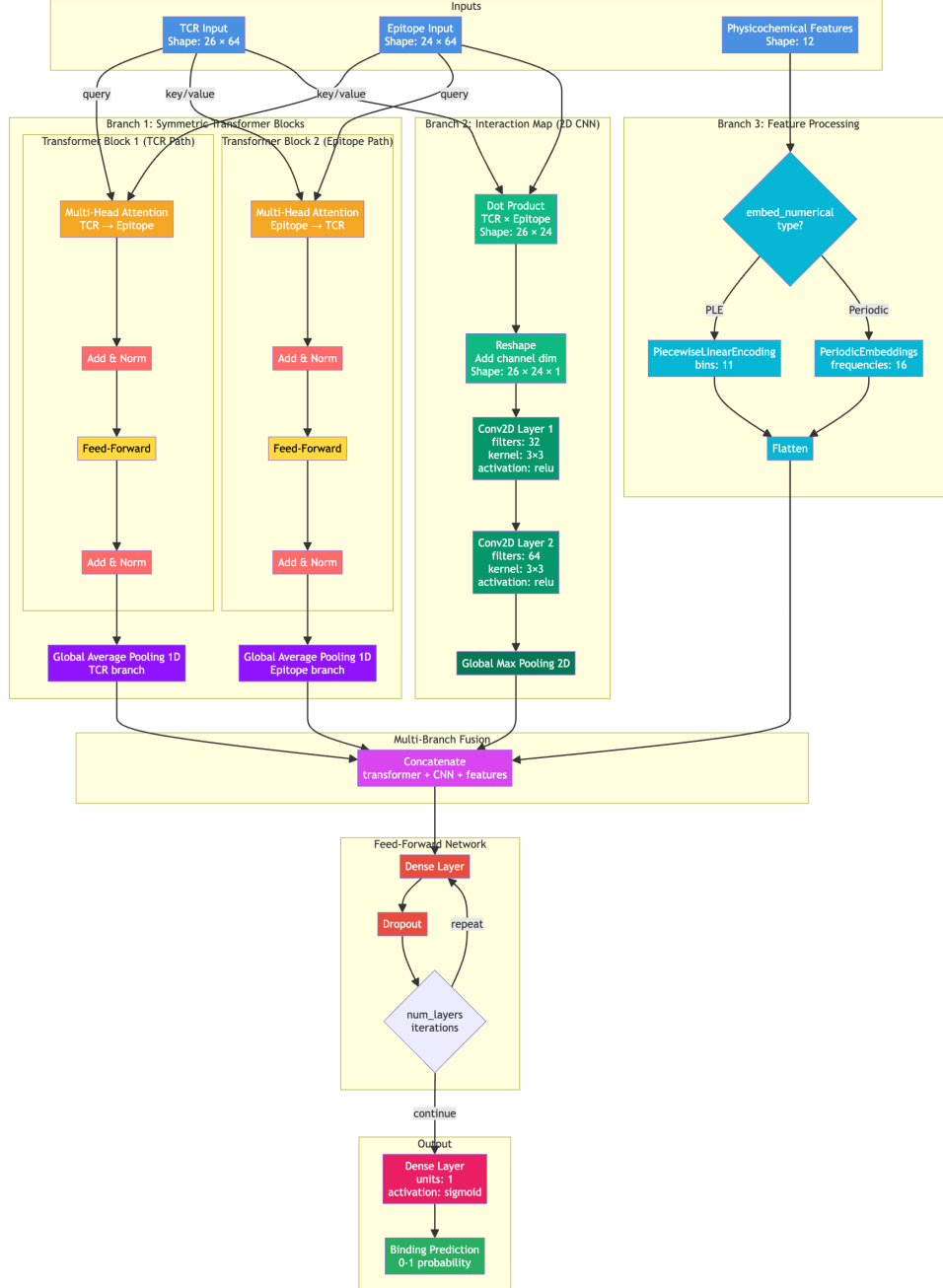


Figure A.6: Architecture of Model Version 6 using Symmetric Cross-Attention, Transformer Blocks and an Interaction Map (2D CNN).

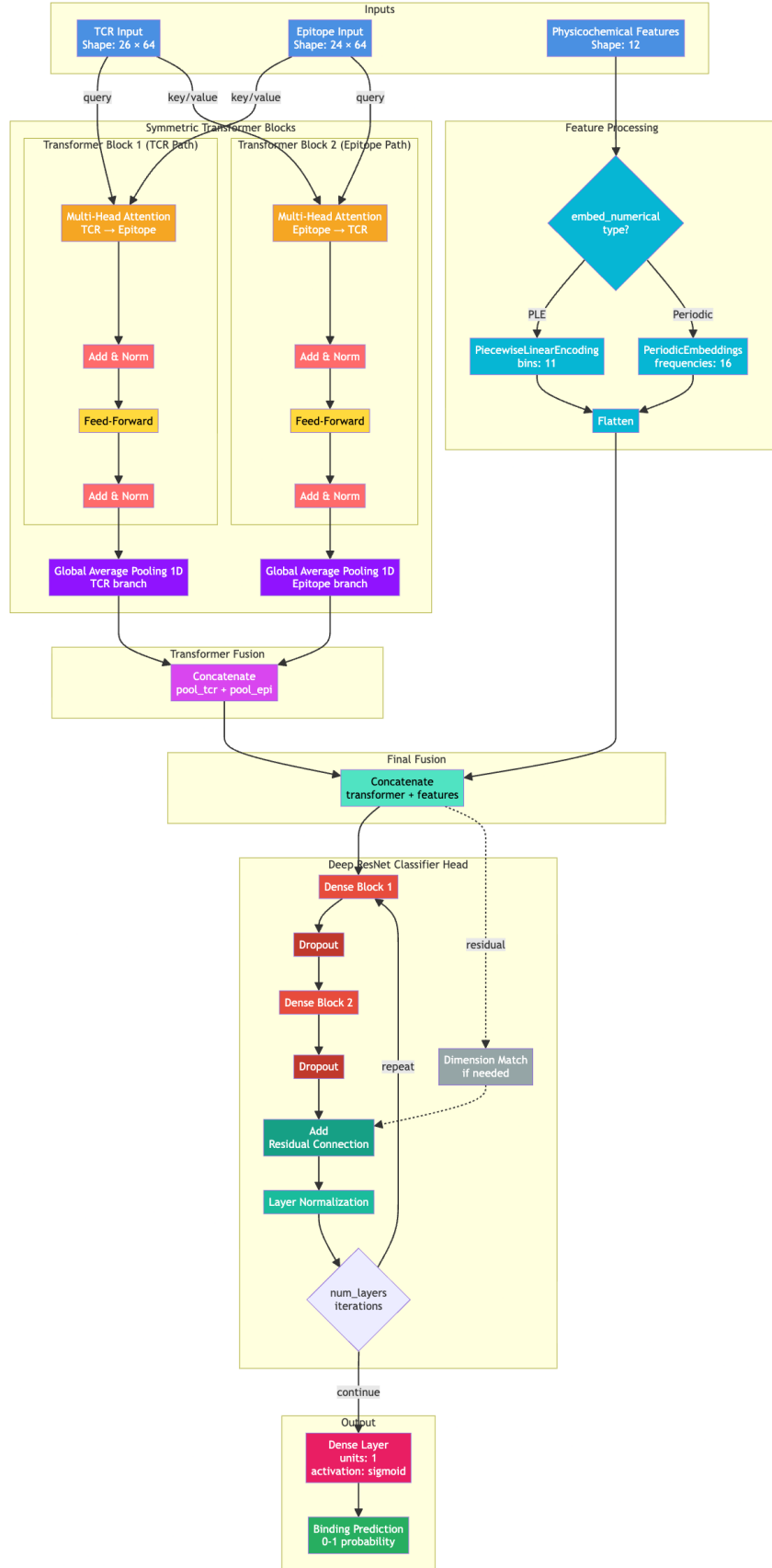


Figure A.7: Architecture of Model Version 7 using Symmetric Cross-Attention, Transformer Blocks and a Deep ResNet Classifier Head.

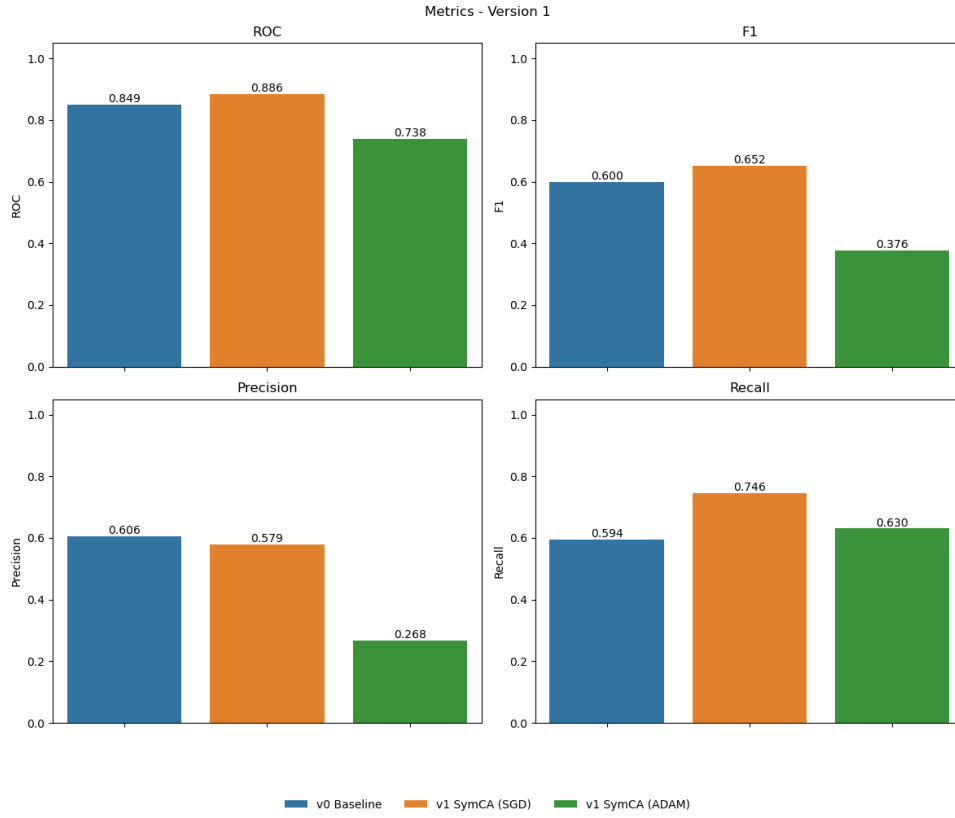


Figure A.8: Model Version 1 using Symmetric Cross-Attention Metrics Plot.

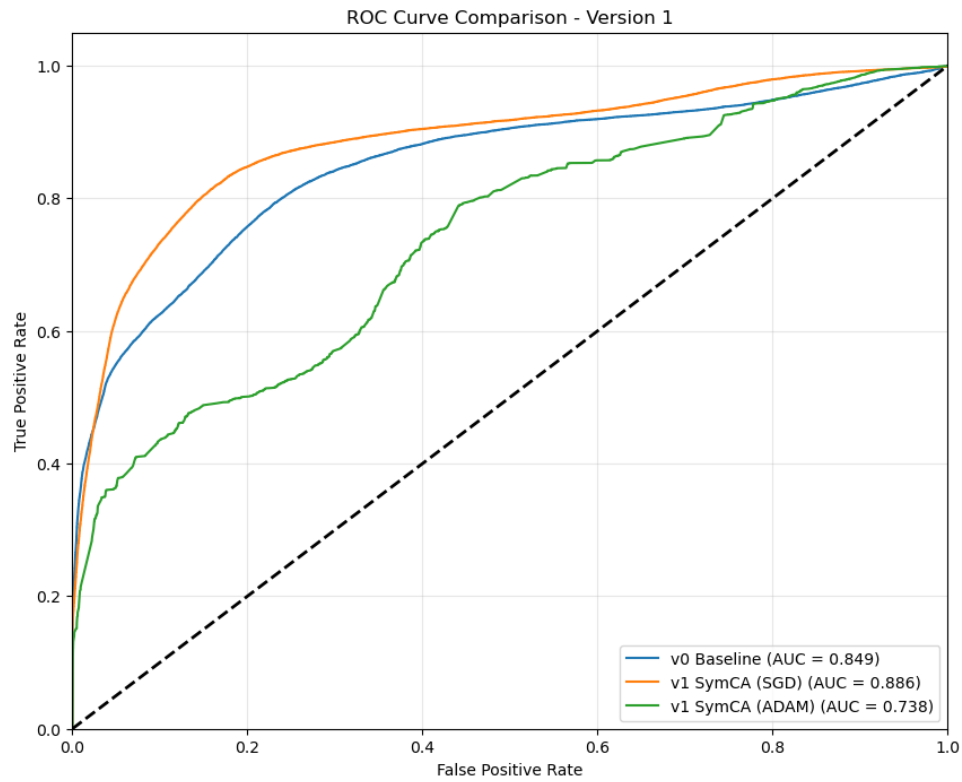


Figure A.9: Model Version 1 using Symmetric Cross-Attention ROC Curves Plot.

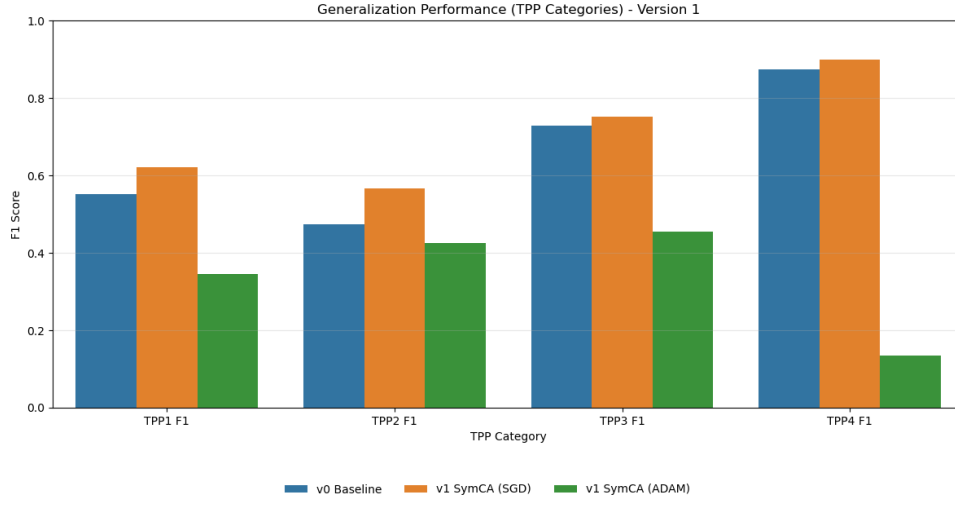


Figure A.10: Model Version 1 using Symmetric Cross-Attention TPP Performance Plot.

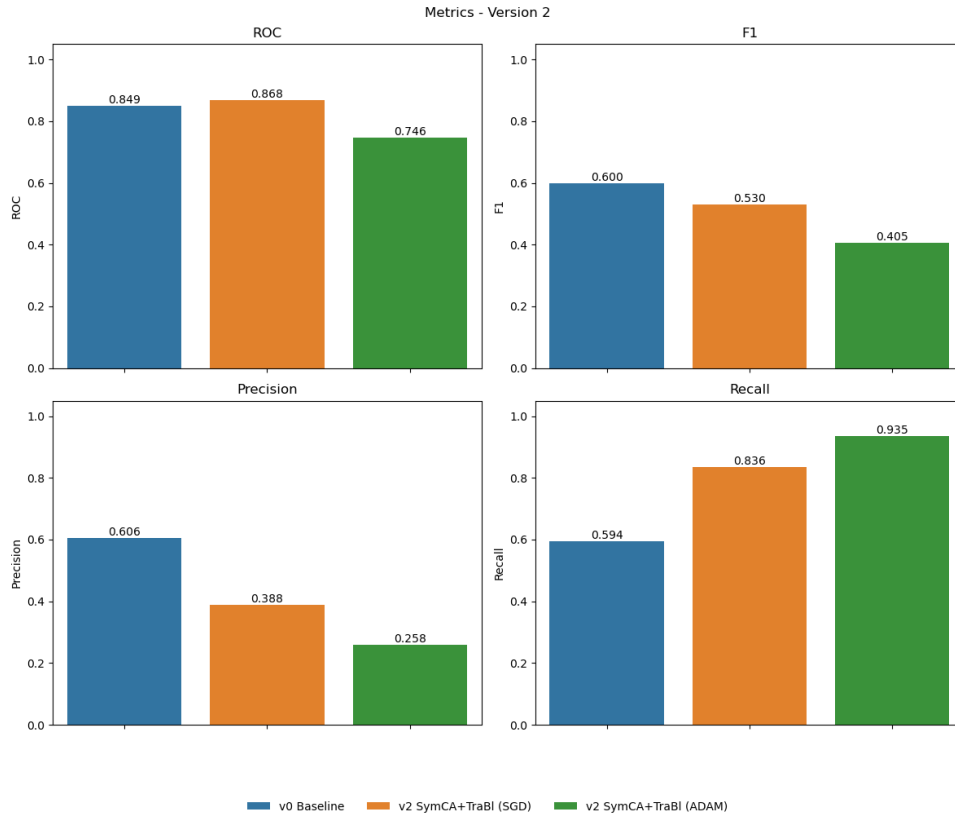


Figure A.11: Model Version 2 using Symmetric Cross-Attention and Transformer Blocks Metrics Plot.

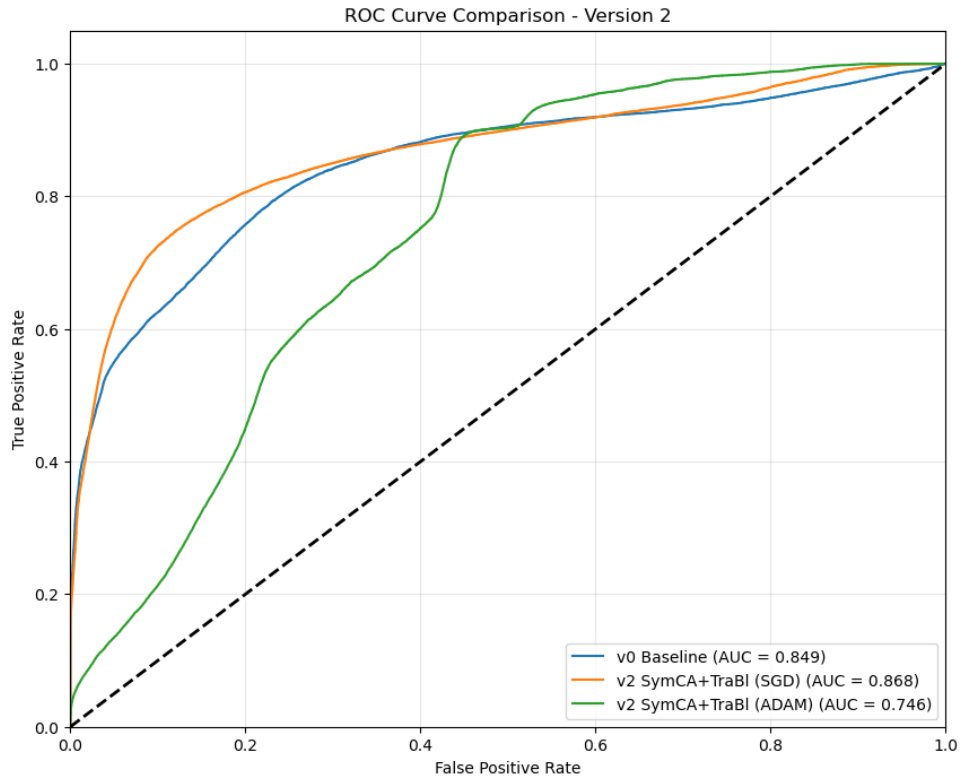


Figure A.12: Model Version 2 using Symmetric Cross-Attention and Transformer Blocks ROC Curves Plot.

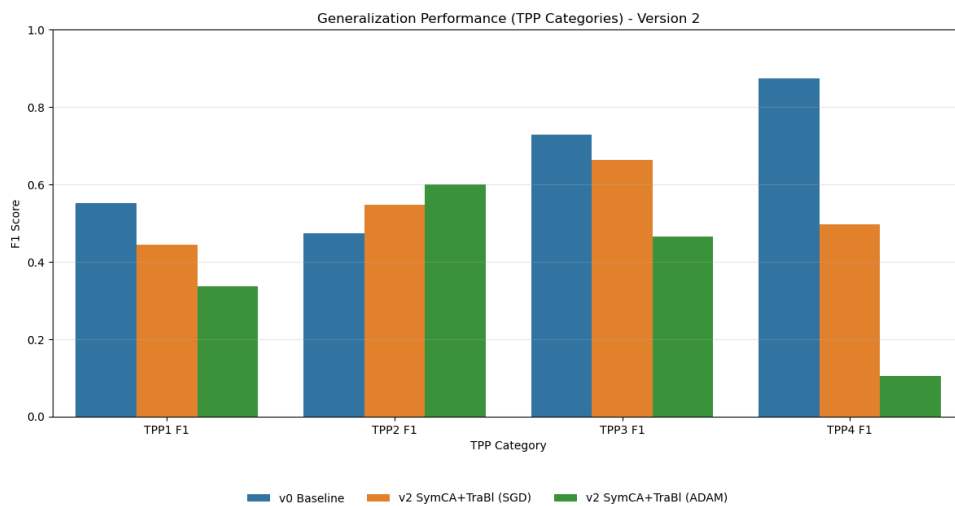


Figure A.13: Model Version 2 using Symmetric Cross-Attention and Transformer Blocks TPP Performance Plot.

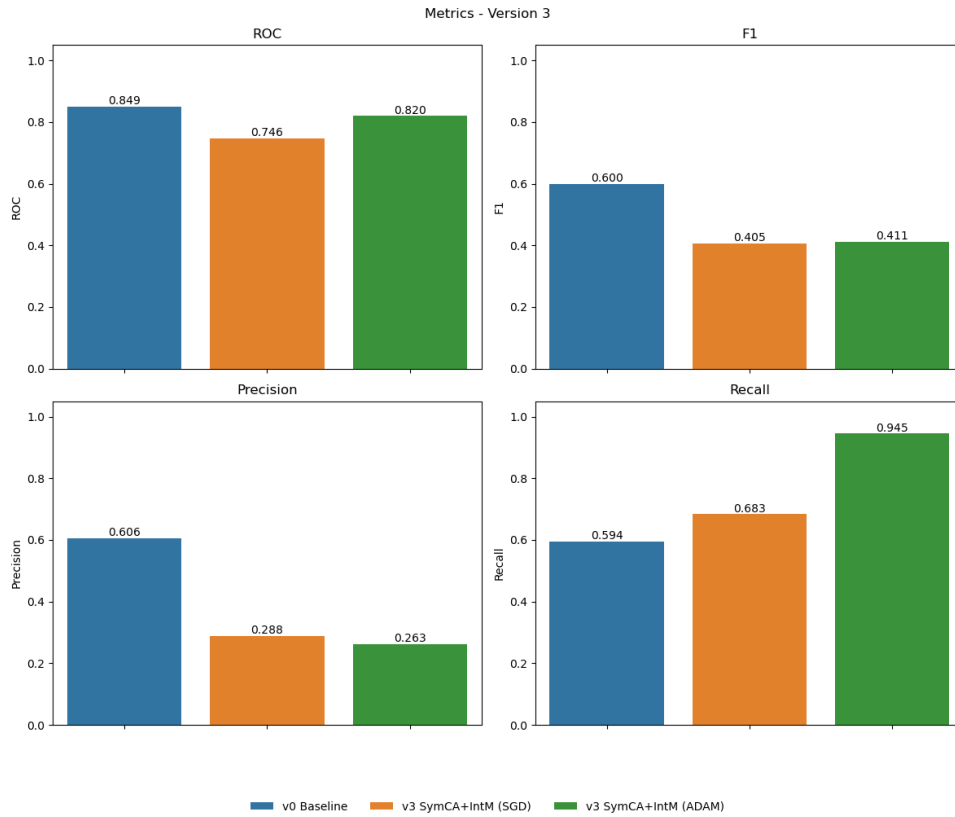


Figure A.14: Model Version 3 using Symmetric Cross-Attention and an Interaction Map (2D CNN) Metrics Plot.

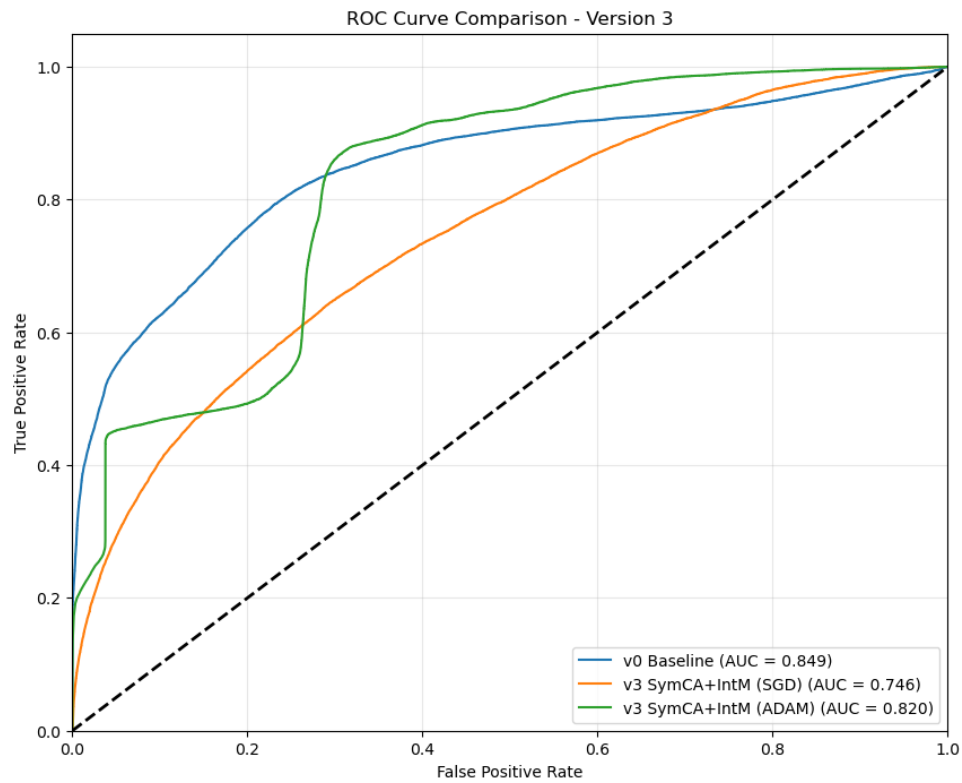


Figure A.15: Model Version 3 using Symmetric Cross-Attention and an Interaction Map (2D CNN) ROC Curves Plot.

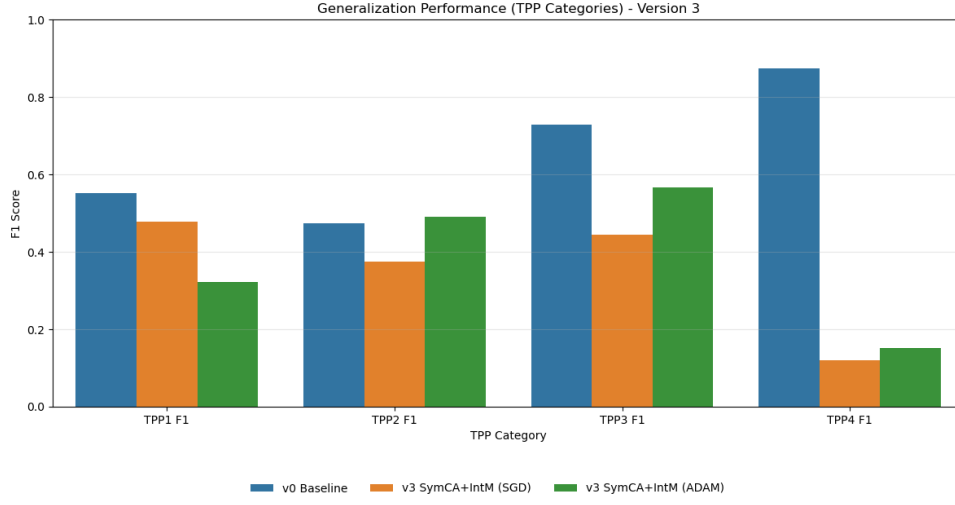


Figure A.16: Model Version 3 using Symmetric Cross-Attention and an Interaction Map (2D CNN) TPP Performance Plot.

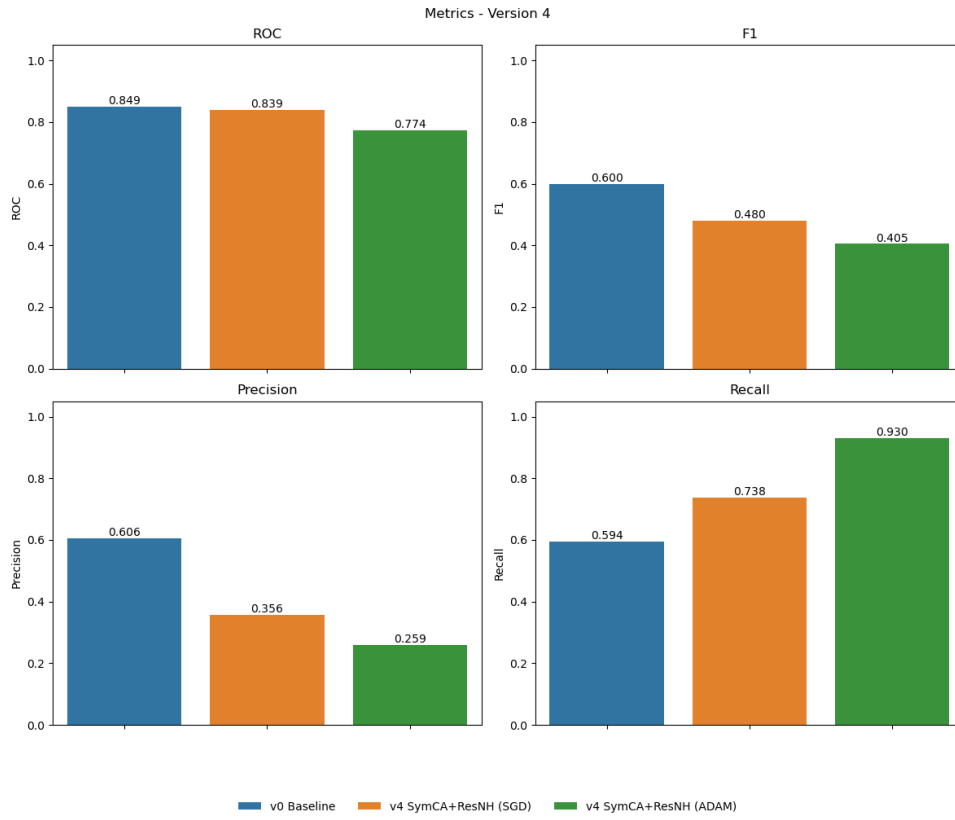


Figure A.17: Model Version 4 using Symmetric Cross-Attention and a Deep ResNet Classifier Head Metrics Plot.

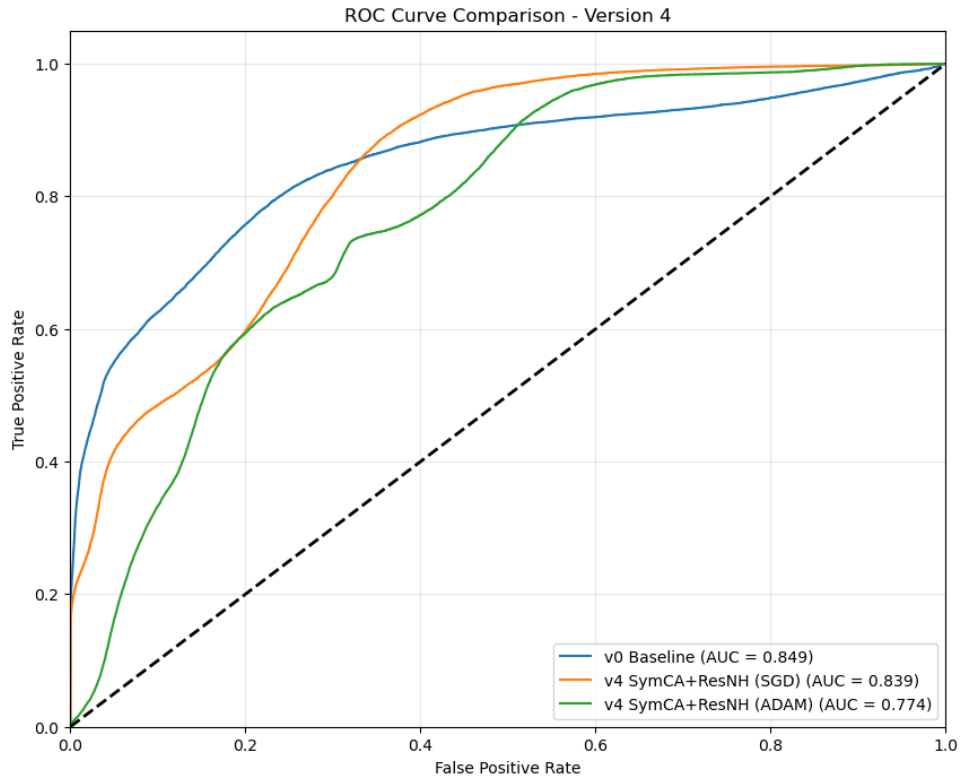


Figure A.18: Model Version 4 using Symmetric Cross-Attention and a Deep ResNet Classifier Head ROC Curves Plot.

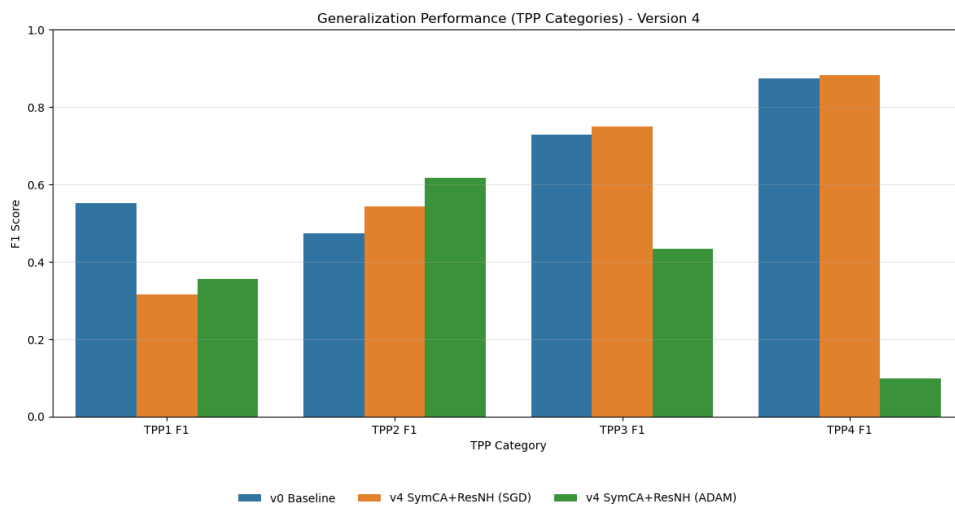


Figure A.19: Model Version 4 using Symmetric Cross-Attention and a Deep ResNet Classifier Head TPP Performance Plot.

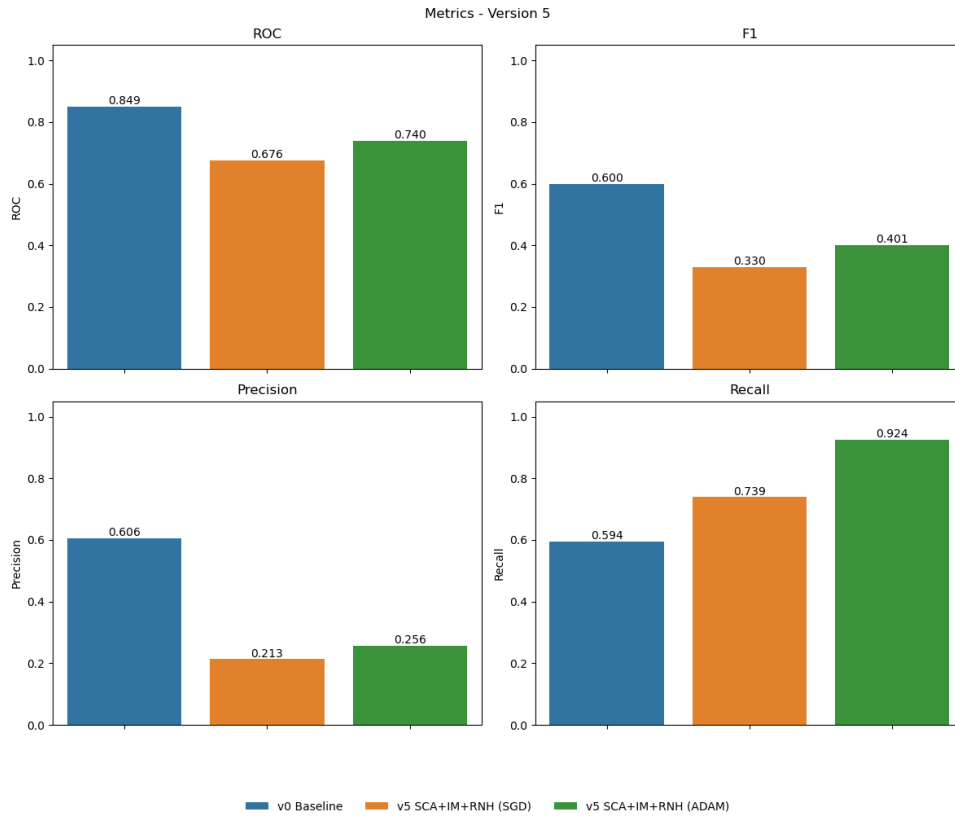


Figure A.20: Model Version 5 using Symmetric Cross-Attention, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head Metrics Plot.

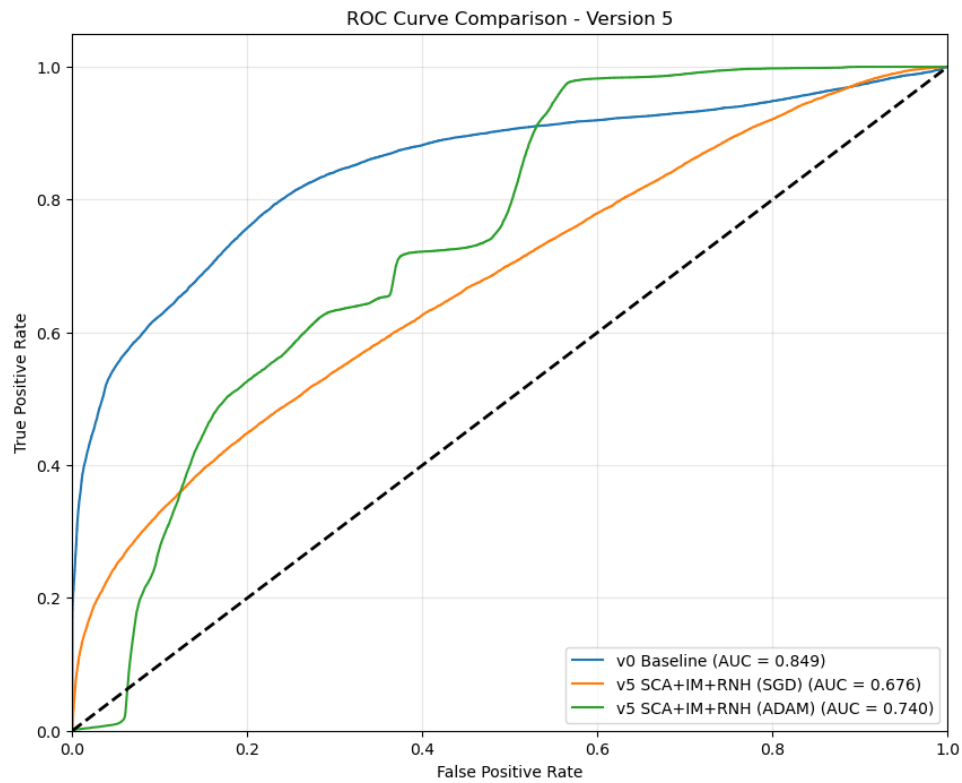


Figure A.21: Model Version 5 using Symmetric Cross-Attention, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head ROC Curves Plot.

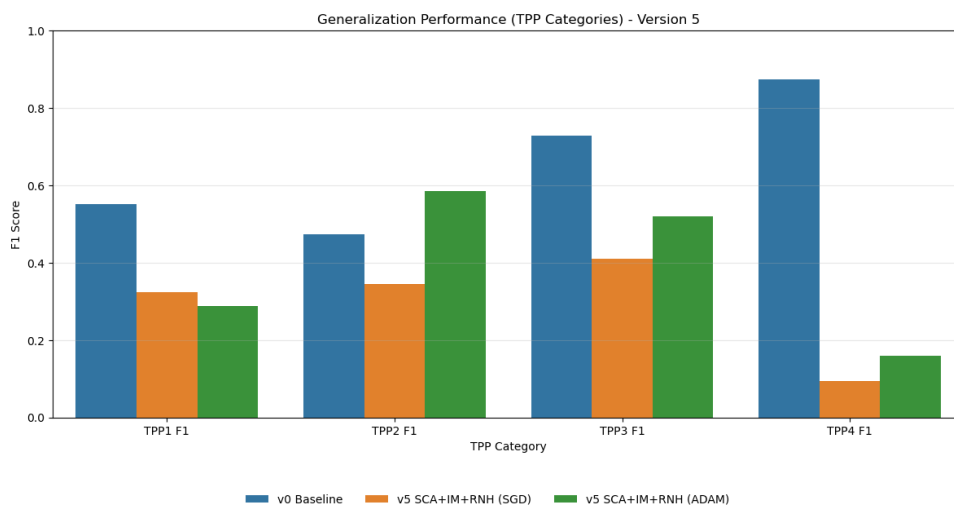


Figure A.22: Model Version 5 using Symmetric Cross-Attention, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head TPP Performance Plot.

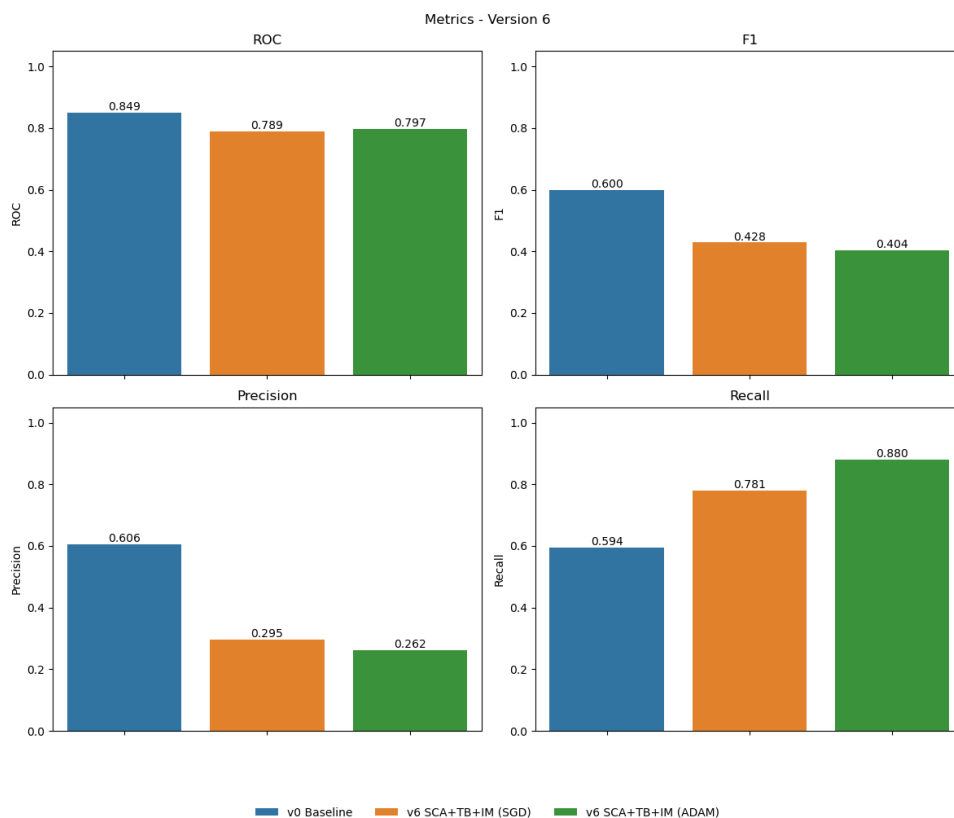


Figure A.23: Model Version 6 using Symmetric Cross-Attention, Transformer Blocks and an Interaction Map (2D CNN) Metrics Plot.

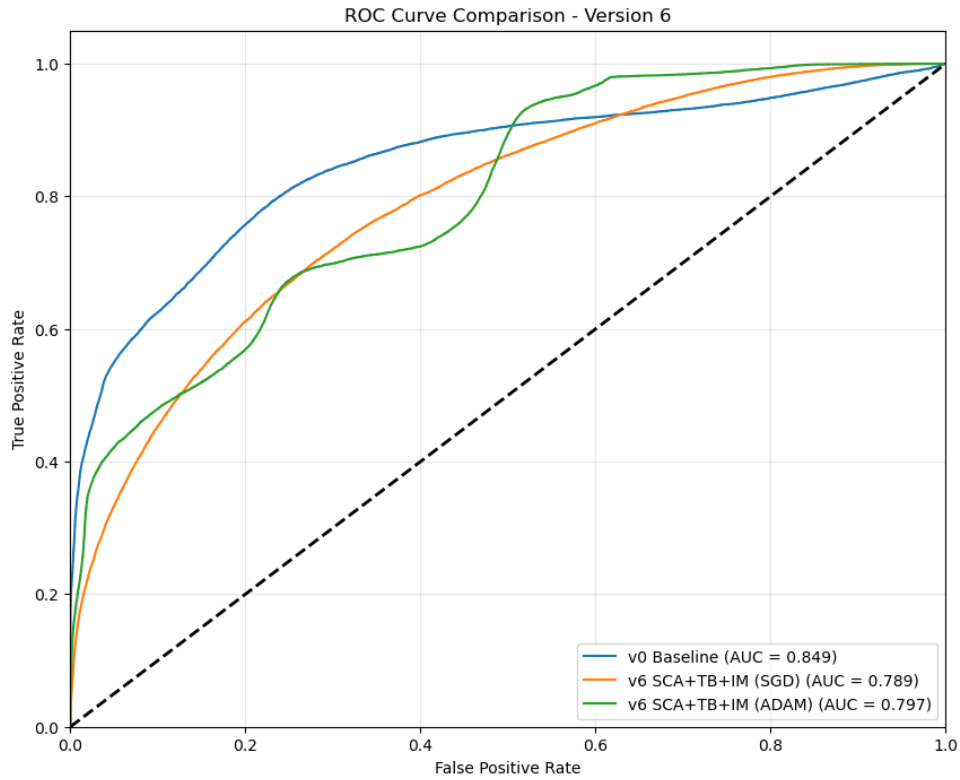


Figure A.24: Model Version 6 using Symmetric Cross-Attention, Transformer Blocks and an Interaction Map (2D CNN) ROC Curves Plot.

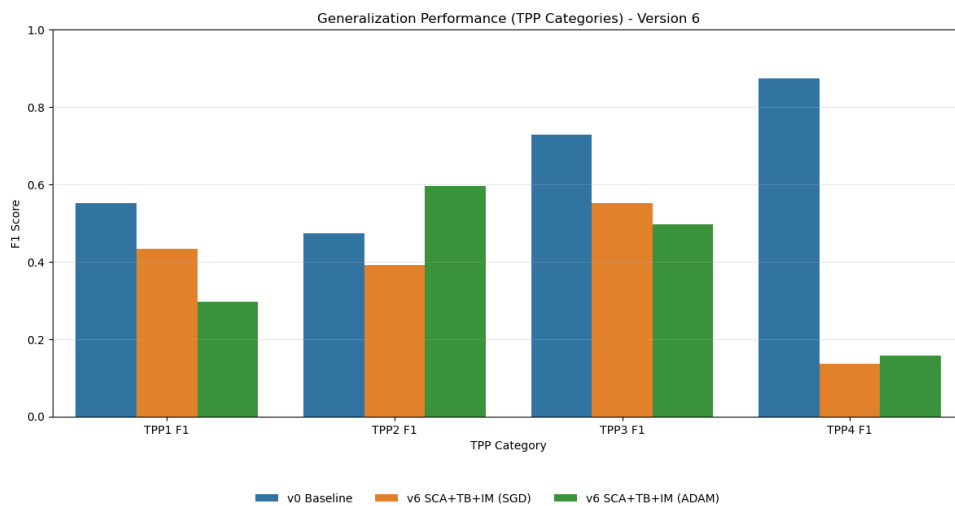


Figure A.25: Model Version 6 using Symmetric Cross-Attention, Transformer Blocks and an Interaction Map (2D CNN) TPP Performance Plot.

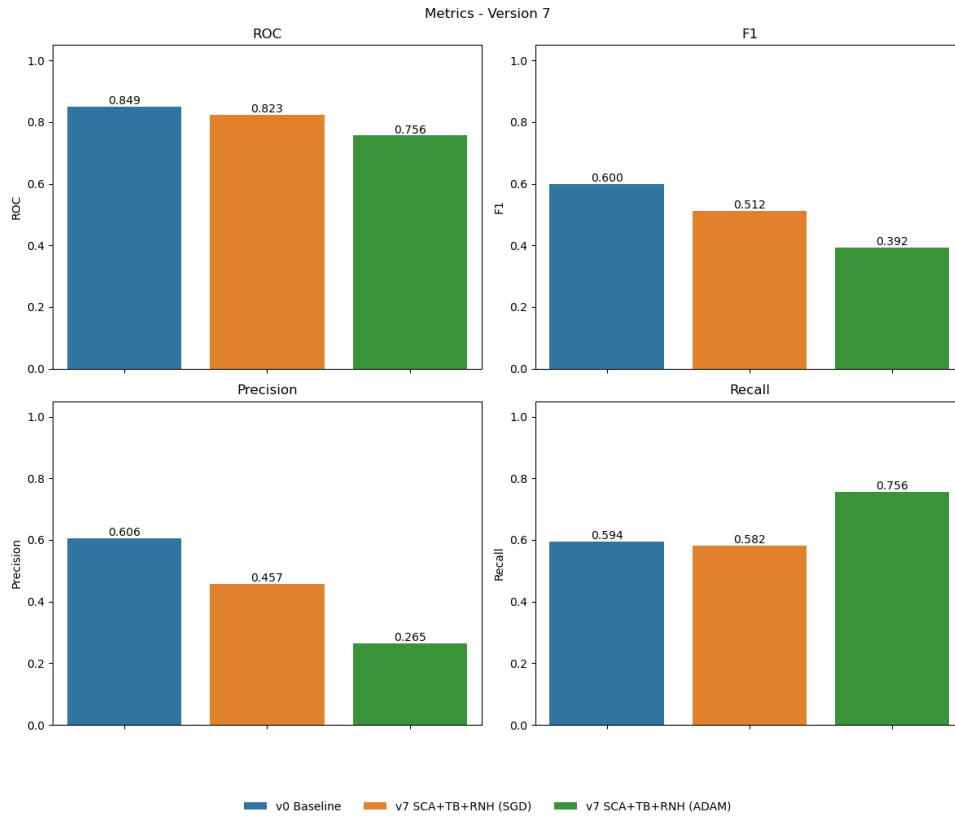


Figure A.26: Model Version 7 using Symmetric Cross-Attention, Transformer Blocks and a Deep ResNet Classifier Head Metrics Plot.

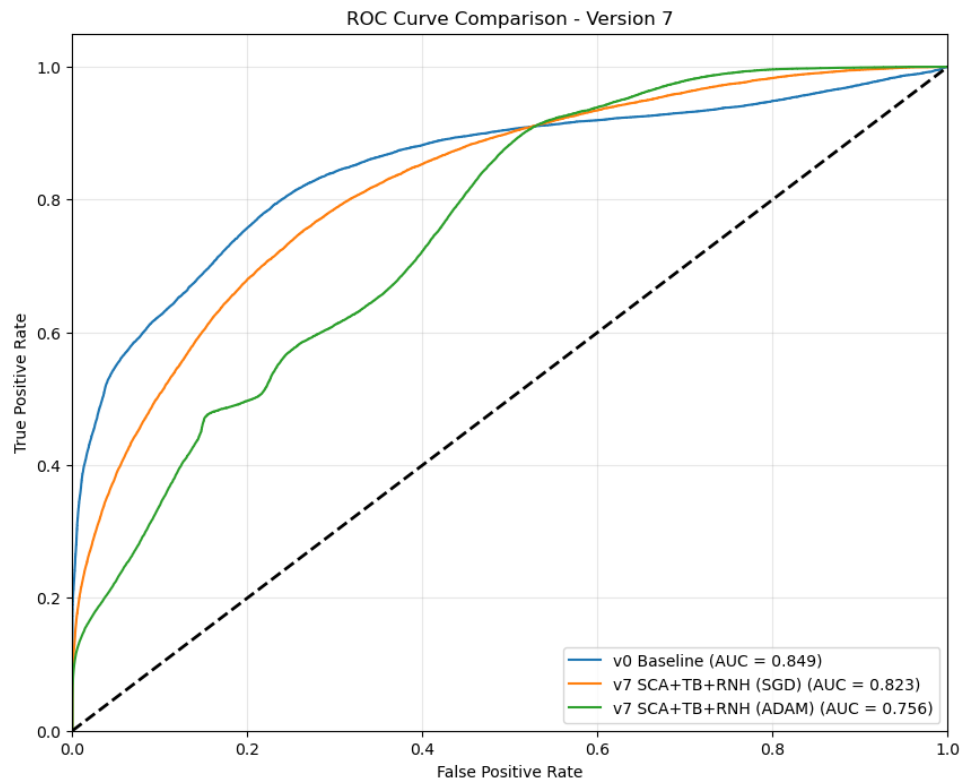


Figure A.27: Model Version 7 using Symmetric Cross-Attention, Transformer Blocks and a Deep ResNet Classifier Head ROC Curves Plot.

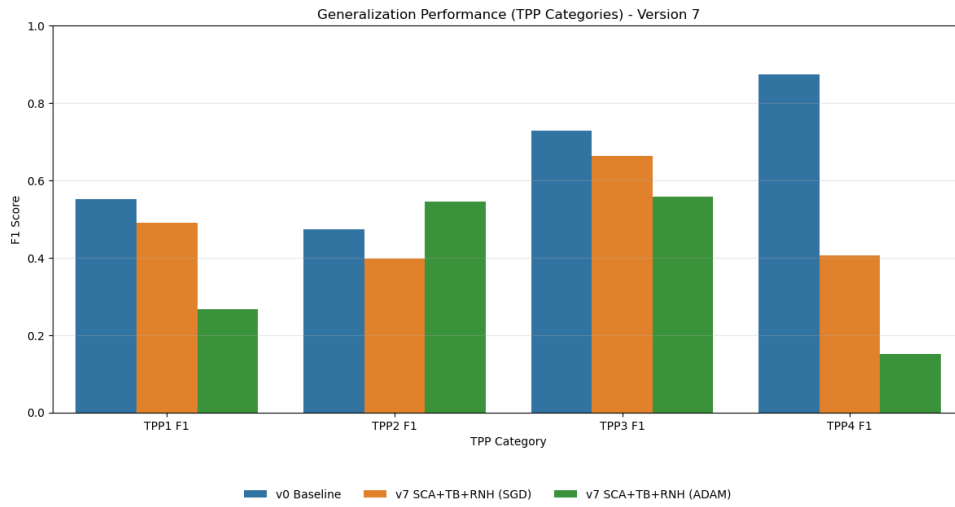


Figure A.28: Model Version 7 using Symmetric Cross-Attention, Transformer Blocks and a Deep ResNet Classifier Head TPP Performance Plot.

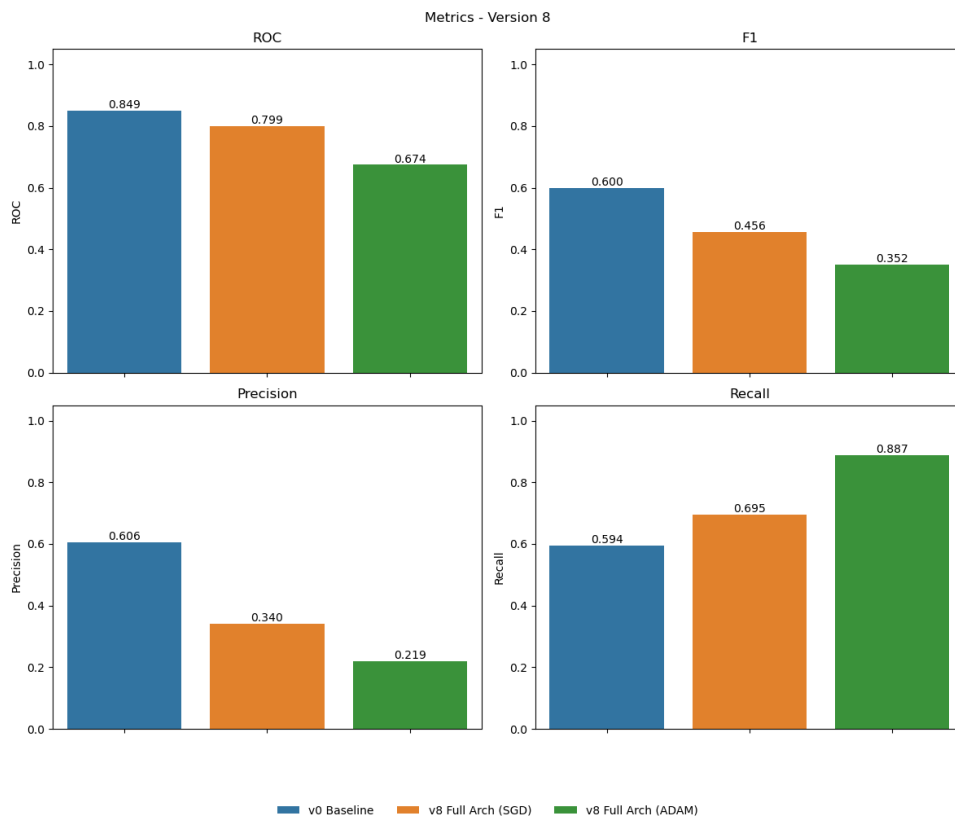


Figure A.29: Model Version 8 using Symmetric Cross-Attention, Transformer Blocks, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head Metrics Plot.

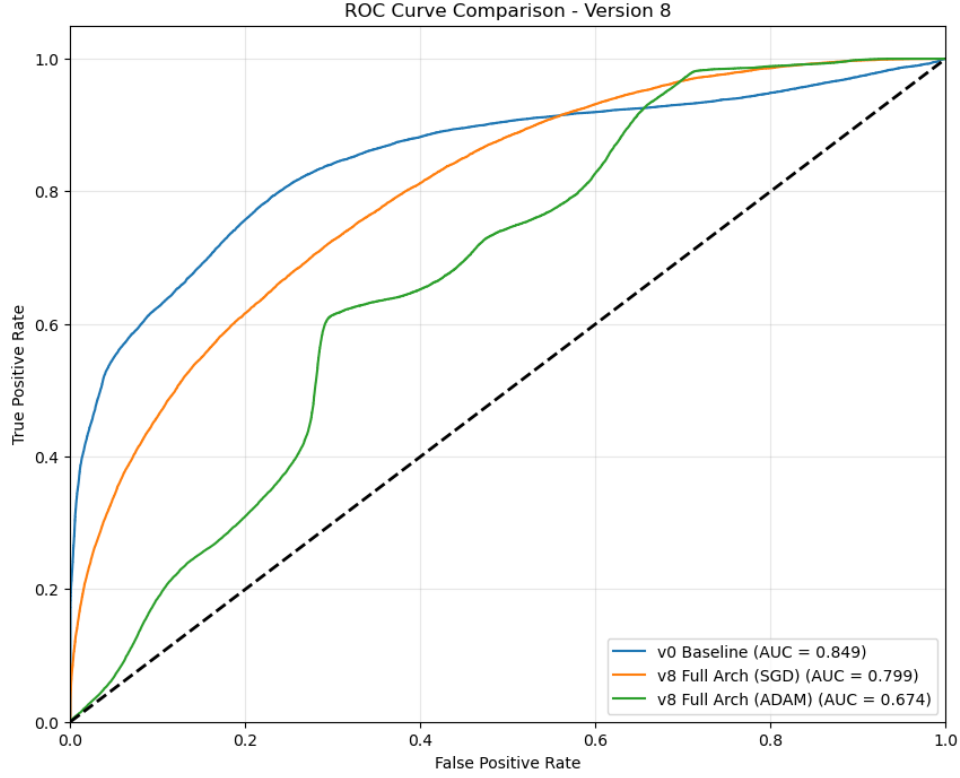


Figure A.30: Model Version 8 using Symmetric Cross-Attention, Transformer Blocks, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head ROC Curves Plot.

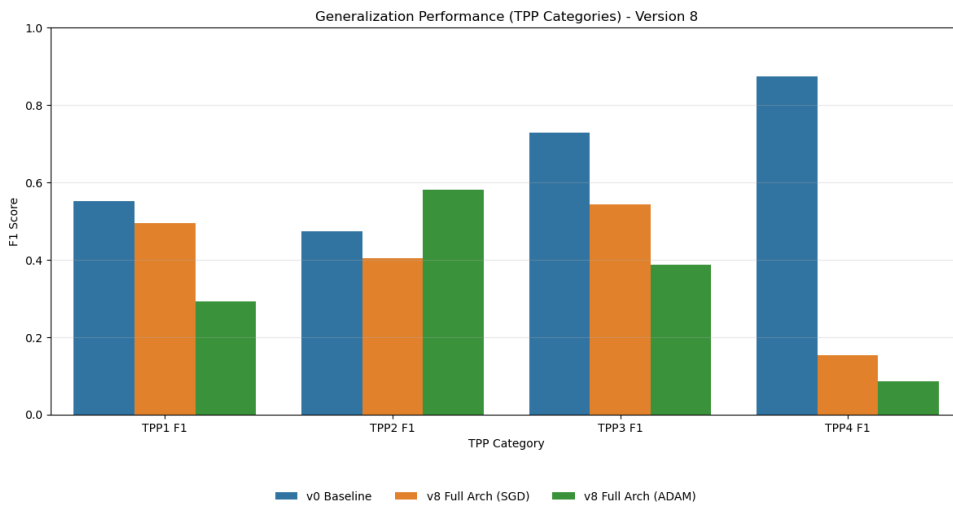


Figure A.31: Model Version 8 using Symmetric Cross-Attention, Transformer Blocks, an Interaction Map (2D CNN) and a Deep ResNet Classifier Head TPP Performance Plot.

Appendix B

Custom Machine Learning Jobs Software

To initially streamline the experimental workflow and manage the training of various model architectures, a custom web-based application titled the "ML Training Job Queue System" was developed. The objective was to abstract away the complexity of environment management and resource allocation.

System Architecture and Features

The application utilizes a containerized microservices architecture:

- **Frontend:** A React-based dashboard allows users to upload jobs, select compute resources (CPU or specific GPUs), and monitor training progress via real-time console log streaming.
- **Backend & Processing:** A FastAPI backend manages API requests, while a background worker utilizes Redis for job queuing. Actual training execution occurs inside isolated Docker containers (utilizing the NVIDIA Container Toolkit), ensuring consistent environments across different runs (e.g., specific PyTorch or TensorFlow versions).
- **Job Structure:** Training jobs are packaged as ZIP files containing the entry script (`main.py`), a dependency file (`requirements.txt`), and the dataset.

Transition to SSH Workflow

While the tool successfully provided a reproducible and user-friendly interface for early experiments, it required the dataset to be included in every uploaded job package. As the complexity of the project grew, repeatedly bundling and uploading large datasets became inefficient. Consequently, the workflow was shifted to connecting directly to the training server via SSH.

Code Availability: The source code for this tool is available on GitHub at: <https://github.com/mvoem/training-queue>.